Bayesian Specification in FAVARs

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Abstract

Every factor-augmented vector autoregression (FAVAR) rests on two choices, which series enter the VAR as observed factors and how many latent factors describe the rest of the panel. Both are usually made by convention. We show that they can be made jointly by marginal likelihood, because every such choice describes the same data, and we give a procedure that makes the comparison feasible in large panels. The comparison requires hierarchical priors on the loadings and on the noise variances. In simulations the procedure recovers the true specification reliably and errs mainly when an observed factor is weak. In a monetary FAVAR on US data it selects eight observed factors, including the three conventional ones, yet the responses to a policy shock are close to those of the conventional specification.

Keywords: factor-augmented VAR, Bayesian model comparison, marginal likelihood, observed factors, number of factors, hierarchical priors.
JEL codes: C11, C32, C38, C52, E52.

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1 Introduction

Economists who study monetary policy want to know how output, prices and employment respond when the central bank raises its interest rate. The standard tool is a vector autoregression (VAR), which models a handful of series together as functions of their own past values. A central bank, however, watches hundreds of series, and a VAR with a handful of them can leave out information that matters. Factor-augmented vector autoregressions (FAVARs), introduced by Bernanke et al. (2005), address this problem. They assume that a large panel of series is driven by a small number of common factors. Most of these factors are latent, which means that they are not observed and must be estimated from the panel. A few are observed series, such as the federal funds rate, that enter the VAR directly. FAVARs are now a standard tool for measuring the effects of monetary policy and other shocks (Boivin et al., 2009; Forni and Gambetti, 2010; Stock and Watson, 2016).

Every FAVAR rests on a decision that is rarely examined, namely which series to treat as observed factors. Bernanke et al. (2005) report results with the federal funds rate as the only observed factor, and with industrial production, the consumer price index and the federal funds rate together. They choose between these two specifications informally, by asking which one gives more plausible impulse responses, and most later applications use one of the two. The number of latent factors is chosen in the same informal way. Bernanke et al. (2005) report results with three and with five. The two choices interact, because a series moved into the VAR can take the place of a latent factor.

The choice matters for three reasons. First, the policy shock is defined relative to the other variables in the VAR, so changing those variables changes the shock. Second, a series that enters the VAR is no longer used to measure the latent factors. Third, treating a series as an observed factor is a strong assumption. It says that the series is a factor measured without error. Bai and Ng (2006) find that industrial production, inflation and unemployment are not exact factors of the Stock and Watson panel, and that the latent factors explain only about one half of the variation in industrial production.

No existing method makes this choice. The closest work asks whether a given observed series is a factor. Bai and Ng (2006) test whether an observed series is an exact linear combination of the latent factors, Andreou et al. (2025) test whether latent and observed factors span the same space, Gagliardini et al. (2019) test whether latent factors remain in the residuals of a model with observed factors, and Bailey et al. (2021) measure the strength of observed factors. These are frequentist tests for static factor models that examine one series or one set of series at a time, so they neither compare all candidate splits jointly nor deliver the VAR and the identified shocks for which a FAVAR is built. Within the FAVAR

literature, Han (2015) tests over-identifying restrictions and Forni and Gambetti (2014) test whether a VAR contains enough information to recover the structural shocks. Kaufmann and Schumacher (2017, 2019), Beyeler and Kaufmann (2021) and Daniele and Schnaitmann (2026) use sparsity in the loadings to determine which series load on which factor, Chao et al. (2023) select the series that are relevant for estimating the factors, Belviso and Milani (2006) give the factors an economic interpretation by construction, and Koop and Korobilis (2014) average over FAVARs that differ in the series used to build the factors. Criteria for the number of factors (Bai and Ng, 2002) and Bayesian comparisons of factor models (Chib et al., 2006) take the set of observed variables as given. None of these papers chooses which series enter the VAR as observed factors, and none chooses them together with the number of latent factors. The problem closest in form arises in asset pricing, where Barillas and Shanken (2018), Chib et al. (2020), Chib and Zeng (2020) and Bryzgalova et al. (2023) compute posterior probabilities over subsets of candidate factors. There the candidates are used to price assets. In a FAVAR they are also part of the panel whose comovement the latent factors describe, and a method for FAVARs has to handle this link between the VAR and the panel.

This paper treats the specification of a FAVAR as a problem of Bayesian model comparison. A specification is the set of series in the VAR together with the number of latent factors. Bayesian statistics compares models by their marginal likelihood, which is the probability of the observed data under a model after averaging over all the parameter values that the prior allows. A model that fits well with plausible parameter values receives a high marginal likelihood, and a model that needs many parameters to fit receives an automatic penalty. We report differences in log marginal likelihoods, which we call log points. A difference of five log points means that the data are about 150 times more likely under one model than under the other, which is conventionally regarded as very strong evidence (Kass and Raftery, 1995). We make five contributions.

The first is to show that the comparison is valid. This is not obvious. Two specifications can assign the same series different roles, one placing it in the VAR and another treating it as a noisy measure of the factors, and models are usually compared only when they explain the same data. We show that they do. Moving a series into the VAR, or changing the number of latent factors, changes how the joint distribution of the series is modelled but not the series themselves. The marginal likelihoods of different specifications are therefore probabilities of the same data and can be compared directly. No series has to be fixed in the VAR in advance, and the data can even decide whether the policy rate is an observed factor. For applied work the output is a posterior probability for each specification, the probability that each candidate series belongs in the VAR, the posterior distribution of the number of

latent factors, and impulse responses averaged over specifications. Marginal likelihoods are the standard way to compare models (Kass and Raftery, 1995), methods to compute them for models with latent variables are well developed (Chib, 1995; Chib and Jeliazkov, 2001), and in factor models they have been used to choose the number of factors (Chib et al., 2006; Chan et al., 2018). They have not been used to choose the split between observed and latent factors.

The second contribution concerns the priors, which matter for the comparison in two places. The first place is the loadings, the coefficients that link each panel series to the factors. Moving a series into the VAR adds a loading on it for every series left in the panel. If each loading has a Gaussian prior with a fixed variance, the log Bayes factor between two specifications contains a term proportional to the log of that variance times the number of added loadings (Remark 1). A large prior variance then penalizes larger VARs by an amount that has nothing to do with the data, and a small one conflicts with the data when the true loadings are large. We avoid this with a hierarchical prior, in which the loadings share a common prior variance that is itself estimated from the many loadings in the panel. The second place is the noise variances, the variances of the parts of each series that the factors do not explain. We show that treating a series as an observed factor is the limit of treating it as the main indicator, or anchor, of an additional latent factor whose noise variance goes to zero (Proposition 2). The two specifications then fit the data almost equally well, and the comparison is decided by how much prior probability the second one gives to a very small noise variance. Under the diffuse priors common in practice the result is close to a tie. With a hierarchical prior on the noise variances, a series whose measurement error is negligible next to that of the rest of the panel is classified as an observed factor. This parallels the argument of Chib et al. (2020) that model comparisons in asset pricing require priors that are proper and consistent across models, and it shares the concern of Chan et al. (2018) that inference in factor models should not depend on arbitrary choices such as the scaling and ordering of the variables.

The third contribution is a procedure that makes the comparison feasible for panels of realistic size. Marginal likelihoods of factor models are hard to compute because every period adds new values of the latent factors, so the integral that defines the marginal likelihood has thousands of dimensions. We integrate out the factors and the loadings on the observed factors exactly. This builds on the precision-based methods of Chan and Jeliazkov (2009), which use banded matrices whose cost grows only linearly with the length of the sample (Lemmas 2 and 3). Bridge sampling (Meng and Wong, 1996), a simulation method for computing marginal likelihoods, is then needed only for the remaining parameters. We also show that under the identification restrictions of Bai et al. (2016) the standard sampler

becomes slower as the panel grows (Proposition 3). We remove this problem by drawing the factors together with the loadings on the observed factors, and by a step that rescales each factor together with its loadings. Once the number of latent factors is free, there are too many specifications to estimate them all. We therefore first form a short list of candidate series, those that principal components explain almost completely. Under conditions we state, this list contains the true observed factors with probability approaching one in large panels (Proposition 5). A local search over the series in the VAR and the number of latent factors then estimates each specification it visits, in parallel. With 100 series a search visits about seventy specifications and takes about six core hours, or two hours on three cores of a laptop. All of these computations carry over to idiosyncratic errors that follow first-order autoregressions, which macroeconomic panels require, at about twice the cost.

The fourth contribution is simulation evidence for panels of the size used in applied work. Bernanke et al. (2005) use 120 monthly series, and the FRED-MD and FRED-QD databases of McCracken and Ng (2016, 2021) contain about 130 and 250 series. We simulate panels of 100 series with three observed and three latent factors, and we start the search with no series in the VAR and no fixed number of factors. The search selects the true specification in all 20 data sets of a baseline design, by at least ten log points. It also selects it in 19 of 20 data sets in which three decoy series are nearly exact linear combinations of the factors. In the exception it prefers a decoy in place of one latent factor, a nearly equivalent specification with fewer parameters. When the dynamics and loadings of one observed factor are scaled down to 30 percent, the true specification is selected in 11 of 20 data sets. In most of the others the weak series is treated as the anchor of an additional latent factor, which is the close comparison that Proposition 2 describes. In designs with one latent factor and the policy rate fixed in the VAR, a fixed-scale prior on the loadings drops an observed factor with weak dynamics in every data set with 100 series, by margins of up to 274 log points. More series do not cure the problem, because the complexity penalty grows with the number of series left in the panel (Appendix A).

The fifth contribution is an application to the monetary FAVAR of Bernanke et al. (2005), with 95 quarterly series from FRED-QD for 1960 to 2007. Two features of the data matter. The first is that the idiosyncratic errors are serially correlated, which means that the part of a series not explained by the factors persists from one quarter to the next. A model that ignores this adds latent factors to absorb the persistence, so we let each error follow its own first-order autoregression. This raises the log marginal likelihood by more than 1600 log points, while on simulated data with serially uncorrelated errors it leaves the margins of the true specification nearly unchanged. The second feature is a limitation of the method. The marginal likelihood keeps rising with the number of latent factors up to at least twelve, and

with eight or more latent factors our estimates do not reproduce across independent runs. We therefore fix the number of latent factors at five, the estimate of Bai and Ng (2002), and report four and six as checks. With the federal funds rate kept in the VAR, the search then adds seven series to it. They are industrial production, consumer prices, the unemployment rate, capacity utilization, employment in trade, transportation and utilities, producer prices and real household assets excluding real estate. The same eight series are selected with four latent factors. The specification with industrial production, consumer prices and the funds rate is preferred to the funds rate alone, but the selected specification is preferred to it by 584 log points. When the same anchor series are used in every specification, the selected specification is still preferred, by about 9 log points over its closest competitor and by 509 log points over the conventional one. The responses to a policy shock nonetheless barely differ between these two specifications, while the funds rate alone gives a more persistent fall in output and a smaller fall in prices. In this panel the choice among plausible observed factors matters more for the description of the panel than for the estimated effects of policy.

The rest of the paper is organized as follows. Section 2 introduces the main ideas, presents the model and its extension to serially correlated errors, and then discusses the specification problem, the priors, the computations and the search. Section 3 reports simulation evidence, Section 4 applies the procedure to a monetary FAVAR, and Section 5 concludes. Appendix A reports further simulations on the priors, Appendix B contains the proofs, and Appendix C gives the sampler with serially correlated errors.

2 Model and specification

2.1 The main ideas

This section uses three ideas, which we first describe informally.

A factor model explains many series with a few common sources of variation. Suppose we observe 100 macroeconomic series every quarter. Many of them rise and fall together over the business cycle, so a few common factors may account for most of their movement. Each series is modelled as a weighted sum of the factors plus a noise term specific to that series. The weights are called loadings, and the noise is called the idiosyncratic error. Principal components, the usual way to estimate such a model, are the weighted averages of the series that capture as much of their joint variation as possible.

A vector autoregression (VAR) describes how a small set of variables evolves over time. Each variable is regressed on past values of all of them. The errors of these regressions, called innovations, are the parts of the variables that could not have been predicted from the

past. A policy shock is built from the innovations. In the recursive scheme that we use, the policy shock is the part of the innovation in the federal funds rate that is not explained by the innovations in the other variables of the VAR in the same quarter. An impulse response then traces how each series moves in the quarters after such a shock. A FAVAR combines the two models. The factors follow a VAR, and the series of the panel load on the factors. Some factors are latent, and others are observed series that enter the VAR directly. This paper asks which series should play that second role.

Bayesian model comparison chooses between models by their marginal likelihood. For a model $\mathcal{M}$ with parameters θ, prior density $\pi(\theta \mid \mathcal{M})$ and likelihood $p(X \mid \theta, \mathcal{M})$, the marginal likelihood is the likelihood averaged over the prior, $m(X \mid \mathcal{M}) = \int p(X \mid \theta, \mathcal{M})\, \pi(\theta \mid \mathcal{M})\, d\theta$. A model that fits the data only for a narrow range of parameter values, or that spreads its prior over many parameters the data do not need, receives a low value. The marginal likelihood therefore rewards fit and penalizes complexity without any further adjustment. When all models have the same prior probability, the posterior probability of each model is proportional to its marginal likelihood. These numbers are tiny, so we work with their logarithms. A difference of d log points means that the data are e^d times more likely under one model than under the other.

The rest of this section makes these ideas precise. Section 2.2 writes down the model, Section 2.3 explains which of its parameters the data can determine, Section 2.4 states the specification problem, and Section 2.5 discusses the priors on which the comparison depends. Sections 2.6 and 2.7 explain how we estimate one specification and its marginal likelihood, and Section 2.8 describes how we search over many specifications.

2.2 The model

Let X_t be an $N \times 1$ vector of standardized series observed for $t = 1, \ldots, T$. A specification assigns a subset S of these series, with $|S| = r_2$, to the VAR as observed factors, $y_t = X_{S,t}$. The remaining $n = N - r_2$ series stay in the panel, $Z_t = X_{-S,t}$. Given r_1 latent factors f_t, write $s_t = (f_t', y_t')'$ for the vector of all factors and $r = r_1 + r_2$ for their number. The FAVAR is

$$Z_t = \Lambda f_t + \Gamma y_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

$$s_t = \sum_{p=1}^{P} B_p s_{t-p} + u_t, \qquad u_t \sim \mathcal{N}(0, \Omega), \tag{2}$$

where $\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ and the errors ε_t and u_t are independent of each other. Equation (1) says that each panel series is a weighted sum of the latent factors, with loadings Λ, and of

the observed factors, with loadings Γ, plus an idiosyncratic error with variance σ_i^2. Equation (2) is a VAR with P lags for all the factors together. In the baseline model both errors are serially uncorrelated. The initial state is $s_1 \sim \mathcal{N}(0, V_0)$, and values of s_t before the sample are set to zero, their mean for standardized series, so that for $t \leq P$ the VAR uses only the lags observed in the sample. The set S may be empty. In that case $r_2 = 0$, Γ is absent and the model is a dynamic factor model. We partition Ω conformably with s_t into Ω_{ff}, Ω_{fy} and Ω_{yy}, and write Λ_1 and Γ_1 for the first r_1 rows of Λ and Γ, and Λ_2 for the remaining rows of Λ. A specification is the triple $\mathcal{M} = (S, r_1, P)$, and $\theta = (\Lambda, \Gamma, \Sigma, B_1, \ldots, B_P, \Omega)$ collects its parameters.

For the stacked system, Z, F and Y are the $T \times n$, $T \times r_1$ and $T \times r_2$ matrices with rows Z_t', f_t' and y_t', and $z = \mathrm{vec}(Z')$, $f = \mathrm{vec}(F')$ and $y = \mathrm{vec}(Y')$ stack the time-t blocks in order. We also write $\tilde{Z}_t = Z_t - \Gamma y_t$ for the panel after removing the part explained by the observed factors, $\tilde{z} = \mathrm{vec}(\tilde{Z}')$, and $G = \Lambda' \Sigma^{-1} \Lambda$.

Serially correlated errors. The idiosyncratic errors of macroeconomic panels are often persistent (Section 4.2). A model with serially uncorrelated errors can account for that persistence only through additional factors. We therefore also consider errors that follow independent first-order autoregressions,

$$\varepsilon_{it} = \rho_i \varepsilon_{i,t-1} + \eta_{it}, \qquad \eta_{it} \sim \mathcal{N}(0, \sigma_i^2), \qquad \varepsilon_{i1} \sim \mathcal{N}\big(0, \sigma_i^2/(1-\rho_i^2)\big), \qquad |\rho_i| < 1, \quad (3)$$

independently across series and of the VAR innovations. In words, the part of series i that the factors do not explain carries a share ρ_i of its value into the next period. Then Σ is the covariance matrix of the innovations η_t, the parameter θ also contains $\rho = (\rho_1, \ldots, \rho_n)$, and $\rho = 0$ gives the baseline model. The coefficients belong to the panel series, so a series that moves into the VAR takes its ρ_j with it, and every split still describes the same data. To treat both models together, let R be the precision matrix, the inverse of the covariance matrix, of $\varepsilon = \mathrm{vec}(E')$, where E is the $T \times n$ matrix with rows ε_t'. In the baseline model $R = I_T \otimes \Sigma^{-1}$. Under (3) the transformation of Prais and Winsten (1954), which multiplies ε_{i1} by $(1-\rho_i^2)^{1/2}$ and replaces ε_{it} by $\varepsilon_{it} - \rho_i \varepsilon_{i,t-1}$ for $t \geq 2$, maps the errors of each series to independent $\mathcal{N}(0, \sigma_i^2)$ variables. With $D_\rho = \mathrm{diag}(\rho_1, \ldots, \rho_n)$, the precision R is then block tridiagonal, with diagonal blocks $\Sigma^{-1}(I_n + w_t D_\rho^2)$, where $w_1 = w_T = 0$ and $w_t = 1$ otherwise, and off-diagonal blocks $-\Sigma^{-1} D_\rho$. Its log determinant is $\log|R| = -T \log|\Sigma| + \sum_{i=1}^n \log(1-\rho_i^2)$. The transformations of the factors in Section 2.3 leave the errors unchanged, so the identification results there hold in both models. The results below also hold in both models unless stated otherwise.

2.3 Identification

A factor model has a well-known indeterminacy. The latent factors are not observed, so we can rescale or mix them and adjust the loadings to compensate, and the data look exactly the same. For example, doubling a latent factor and halving its loadings leaves every series unchanged. The data cannot tell such versions of the model apart, so some parameters must be fixed by convention. This step is called normalization, or identification.

Formally, for an invertible $r_1 \times r_1$ matrix M_{11} and any $r_1 \times r_2$ matrix M_{12}, let

$$A = \begin{pmatrix} M_{11} & M_{12} \\ 0 & I_{r_2} \end{pmatrix}, \qquad f_t^* = M_{11} f_t + M_{12} y_t,$$

so that $s_t^* = A s_t$. Setting $\Lambda^* = \Lambda M_{11}^{-1}$, $\Gamma^* = \Gamma - \Lambda M_{11}^{-1} M_{12}$, $B_p^* = A B_p A^{-1}$ and $\Omega^* = A \Omega A'$ gives a model with the same joint distribution of (Z, Y). Identification therefore requires $r_1^2 + r_1 r_2$ restrictions. Bai et al. (2016) consider three sets of restrictions, each of which requires $\Omega_{fy} = 0$, so that the innovations of the latent factors are uncorrelated with those of the observed factors. We use the third set, which we denote IRc. It sets $\Lambda_1 = I_{r_1}$ and leaves Ω_{ff} unrestricted. In words, each of the first r_1 series of the panel loads on exactly one latent factor with a loading of one. These series act as anchors, and each latent factor is measured in the units of its anchor. Which series are ordered first is part of the normalization and is discussed below. Bernanke et al. (2005) instead impose $\Lambda_1 = I_{r_1}$ and $\Gamma_1 = 0$ and leave Ω unrestricted. We refer to this normalization as BBE.

Proposition 1 (Observational equivalence)**.** *Let θ satisfy IRc with Ω_{yy} positive definite, and apply the transformation above with $M_{11} = I_{r_1}$ and $M_{12} = \Gamma_1$. The transformed parameters satisfy BBE, the transformed factor is $f_t^{\mathrm{BBE}} = f_t + \Gamma_1 y_t$, and the joint distribution of (Z, Y) is unchanged. Conversely, for θ satisfying BBE, the transformation with $M_{11} = I_{r_1}$ and $M_{12} = -\Omega_{fy}\Omega_{yy}^{-1}$ yields parameters that satisfy IRc, and the two maps are inverse to each other. Finally, the common component $\chi_t = \Lambda f_t + \Gamma y_t$ and the responses*

$$R_h = \frac{\partial}{\partial v_t'} \mathrm{E}\left[\begin{pmatrix} Z_{t+h} \\ y_{t+h} \end{pmatrix} \,\middle|\, v_t,\ s_{t-1}, s_{t-2}, \ldots \right], \qquad h \geq 0,$$

of the observed series to the innovation v_t in y_t take the same values under every parameterization in the class above.

Proposition 1 says that IRc and BBE describe the same model in different units, and that the impulse responses of the observed series do not depend on which one is used. This has a direct implication for specification. Within a given split, choosing between exactly

identifying normalizations is a choice of labels for the same model, and the data cannot rank them. The split itself is a different matter, as the next subsection explains.

The choice of anchors within IRc is a normalization of the same kind. Any r_1 series whose loadings on the latent factors form an invertible block can serve as anchors, and the transformation above with $M_{12} = 0$ and M_{11} equal to that block maps one choice into another without changing the likelihood. The priors of Section 2.5, however, are placed on the normalized parameters, so the marginal likelihood is not exactly invariant to the choice. If the anchor block is close to singular, the normalized loadings ΛM_{11}^{-1} of the other series are large, and the prior on the loadings penalizes them. We therefore choose the anchors by a rule that is applied in the same way in every specification. We compute r_1 principal components of the panel series after projecting out y, and rotate their loadings by varimax (Kaiser, 1958), a rotation that makes each series load mainly on one component. The anchor of each rotated component is then the series with the largest absolute loading on it among those not yet chosen. The rule selects series that each measure mainly one factor, so the anchor block is well conditioned.[1]

2.4 The specification problem

To see why different specifications can be compared, think of the data as the full table of T observations on N series. Two specifications $\mathcal{M} = (S, r_1, P)$ and $\mathcal{M}' = (S', r_1', P')$ that use the same collection of series describe this same table, $X = (X_1', \dots, X_T')'$. They differ only in how they model it. Each implies a joint density $p(X \mid \theta, \mathcal{M}) = p(Z, Y \mid \theta, \mathcal{M})$ with respect to Lebesgue measure on $\mathbb{R}^{TN}$, so the marginal likelihoods

$$m(X \mid \mathcal{M}) = \int p(X \mid \theta, \mathcal{M})\, \pi(\theta \mid \mathcal{M})\, d\theta$$

are densities of the same random vector and can be compared. Posterior model probabilities $\Pr(\mathcal{M} \mid X) \propto m(X \mid \mathcal{M}) \Pr(\mathcal{M})$ over any finite set of specifications follow in the usual way (Kass and Raftery, 1995). From these we obtain the posterior probability that each candidate series belongs in the VAR, $\Pr(j \in S \mid X) = \sum_{\mathcal{M}: j \in S} \Pr(\mathcal{M} \mid X)$, which we call its inclusion probability. We also obtain posterior distributions of quantities that are defined in every specification, such as the responses of the panel to the policy innovation, averaged over specifications.

[1] In a simulated panel of 30 series with three latent factors, the log marginal likelihood with the anchors chosen by this rule was within 0.6 log points of the value with the true anchors. Choosing as anchors the series that the factors explain best lowered it by 28 log points. Section 4.4 returns to the anchors in the application.

Two versions of the question are of interest. The first holds r_1 fixed. Moving a series from the panel into the VAR then asks whether it carries dynamics of its own that the latent factors do not capture. The second holds the total number of factors $r = r_1 + r_2$ fixed. The comparison between $(S \cup \{j\}, r_1)$ and $(S, r_1 + 1)$ then asks whether series j is itself a factor measured without error, or is better described as a noisy indicator of an additional latent factor. The two specifications have the same number of factors and differ mainly in whether series j is measured with error, so this comparison is close to the question of exact factors in Bai and Ng (2006). It is also the hardest comparison, because an additional latent factor anchored on series j can reproduce an observed factor as closely as its noise variance allows. Section 2.5 makes this precise and draws the consequence for the prior.

Nothing in the comparison requires a particular series to be in the VAR. When the analysis concerns a policy shock, the policy rate can be kept in the VAR in every specification so that the policy innovation is defined in all of them, and the comparison then concerns the other series. The search of Section 2.8 also allows the set of observed factors to be empty, and then the data decide whether the policy rate is an observed factor.

Two comparisons are not available. By Proposition 1, exactly identifying normalizations imply the same likelihood and differ only through their priors. Specifications estimated on panels with different series describe different data.

2.5 Priors that are consistent across specifications

A prior describes what we believe about the parameters before seeing the data. When models are compared by marginal likelihood, the prior matters much more than it does for estimation within one model. The marginal likelihood averages the fit over the prior, so a prior that is very spread out puts little weight on the parameter values that fit well and lowers the marginal likelihood. This effect does not disappear in large samples. The comparison therefore requires proper priors, and it requires that parameters which change dimension across specifications receive priors that do not favour one specification for reasons unrelated to the data. The loadings are the main issue. When series j moves from the panel into the VAR, the panel loses the row (Λ_j, Γ_j), and every remaining series gains a loading on y_j, so Γ gains a column of length $n - 1$.

Remark 1 (Complexity penalty and prior scale). Suppose the free loadings of a specification are $\mathcal{N}(0, vI_k)$ a priori, with k the number of free loadings, and let $\hat{\theta}$ be the posterior mode. A Laplace approximation to the integral over the loadings gives

$$\log m(X \mid \mathcal{M}) \approx \log p(X \mid \hat{\theta}, \mathcal{M}) - \frac{k}{2} \log v - \frac{\|\hat{\lambda}\|^2}{2v} - \frac{1}{2} \log \left| H_\lambda + v^{-1} I_k \right| + c,$$

where $\hat{\lambda}$ collects the loadings at the mode, H_λ is the negative Hessian of the log likelihood with respect to them and c does not depend on v. For two specifications with k and k' free loadings, the log Bayes factor therefore contains the term $-\frac{1}{2}(k-k')\log v$. With a diffuse prior this term penalizes the specification with more loadings by an amount that grows without bound in v. With a tight prior the term $\|\hat{\lambda}\|^2/2v$ dominates and penalizes specifications whose loadings are large, whatever their fit.

The intuition behind Remark 1 is simple. The data pin down each loading to a narrow range. Under a prior with variance v, that range is a share of order $1/\sqrt{v}$ of the prior, so each additional loading lowers the log marginal likelihood by about $\frac{1}{2}\log v$. A fixed-scale prior on the loadings, however it is chosen, therefore introduces an arbitrary component into the comparison of splits. We use a hierarchical prior instead. The free loadings on the latent factors and the loadings on the observed factors are independent Gaussian vectors, each with a common variance,

$$\Lambda_{ij} \mid \tau_\Lambda^2 \sim \mathcal{N}(0, \tau_\Lambda^2), \quad i > r_1, \qquad \Gamma_{ij} \mid \tau_\Gamma^2 \sim \mathcal{N}(0, \tau_\Gamma^2), \qquad \tau_\Lambda^2, \tau_\Gamma^2 \sim \mathcal{IG}(a_\tau, b_\tau),$$

and the same hyperprior, the prior on the common variances, is used in every specification. Integrating over the common variances gives a multivariate t prior on each set of loadings. The common variances, which we also call scales, are estimated from the cross-section of loadings, which has many elements in any realistic panel. The term $\frac{k}{2}\log v$ in Remark 1 is therefore replaced by a quantity determined by the data, and a tight prior cannot conflict with large loadings, because the scale adapts to them. The only choice left to the researcher is (a_τ, b_τ), which is shared by all specifications. We use $a_\tau = b_\tau = 1/2$, and Appendix A.2 reports the sensitivity of the results to this choice.[2]

The idiosyncratic variances also receive a hierarchical prior,

$$\sigma_i^2 \mid s_\sigma \sim \mathcal{IG}(\nu_\sigma/2,\, s_\sigma/2), \quad i = 1, \ldots, n, \qquad s_\sigma \sim \mathcal{G}(a_\sigma, b_\sigma),$$

where $\mathcal{G}(a, b)$ is the gamma distribution with shape a and rate b. Each noise variance is drawn from a distribution whose scale s_σ is shared by all series and estimated from them, so the typical noise variance in the panel informs the prior for each series. The parameter ν_σ sets how tightly each noise variance is pulled toward this typical value. The reason for this

[2]The hierarchical prior on the latent-factor loadings also removes a separate problem of the IRc normalization. Along the path $\Lambda_2 \mapsto c\Lambda_2$, $f \mapsto f/c$, the posterior acquires a volume factor that grows with the number of free loadings. With a fixed-scale prior this factor is balanced only by the prior variance, so the prior variance determines the scale of the factor when the anchor series is noisy.

prior is given at the end of this section. Integrating over s_σ gives the joint prior

$$\pi(\sigma_1^2, \ldots, \sigma_n^2) = \frac{b_\sigma^{a_\sigma}\,\Gamma(a_\sigma + n\nu_\sigma/2)}{\Gamma(a_\sigma)\,\{2^{\nu_\sigma/2}\Gamma(\nu_\sigma/2)\}^n}\;\frac{\prod_{i=1}^n (\sigma_i^2)^{-\nu_\sigma/2-1}}{\left(b_\sigma + \frac{1}{2}\sum_{i=1}^n \sigma_i^{-2}\right)^{a_\sigma + n\nu_\sigma/2}}, \tag{4}$$

which is used in the marginal likelihood, so s_σ adds no dimension to the integral. We use $\nu_\sigma = 1$ and $a_\sigma = b_\sigma = 1$. The other priors are standard and have the same form in every specification. The VAR coefficients are independent Gaussian variables with mean zero and variance b/p^2 for the coefficients on lag p, so that more distant lags are shrunk more toward zero. We do not restrict them to values that make the VAR stationary, since that restriction would introduce a normalizing constant that depends on the dimension of the VAR. Under IRc the innovation covariance is block diagonal with $\Omega_{yy} \sim \mathcal{IW}(r_2 + 2, I_{r_2})$ and $\Omega_{ff} \sim \mathcal{IW}(r_1 + 2, I_{r_1})$, where $\mathcal{IW}$ is the inverse Wishart distribution. Under (3) the autoregressive coefficients are independent with $\rho_i \sim \mathcal{N}(0, 0.5^2)$ truncated to $(-1, 1)$. All series are standardized before estimation, so that a common prior scale is meaningful across series.

Observed factors and nearly exact latent factors. The comparison between $(S \cup \{j\}, r_1)$ and $(S, r_1 + 1)$ in Section 2.4 has a special structure. The next result shows that the first specification lies on the edge of the second. In other words, a model with series j as an observed factor is what a model with one more latent factor becomes when the noise in series j shrinks to zero.

Proposition 2 (An observed factor as the limit of a latent factor)**.** *Let $\mathcal{M} = (S \cup \{j\}, r_1, P)$ and $\mathcal{M}' = (S, r_1 + 1, P)$ with $j \notin S$, and let θ be a parameter of $\mathcal{M}$ that satisfies IRc with Ω_{yy} positive definite. There is a parameter θ_0' with the structure of $\mathcal{M}'$ that satisfies IRc, with series j as the anchor of the additional latent factor, and that differs from a parameter of $\mathcal{M}'$ only in that $\sigma_j^2 = 0$, such that the following holds. Let θ_ϵ' be θ_0' with σ_j^2 replaced by ϵ. Then θ_ϵ' is a parameter of $\mathcal{M}'$ for every $\epsilon > 0$, and $p(X \mid \theta_\epsilon', \mathcal{M}') \to p(X \mid \theta, \mathcal{M})$ for every X as $\epsilon \downarrow 0$. As in Proposition 1, the covariance of the initial state transforms together with the states.*

Proposition 2 explains why this comparison is hard. A specification that treats an observed factor as an additional latent factor can come arbitrarily close to the true distribution of the data, and it does so by letting the noise variance of the series go to zero. The likelihood alone therefore separates the two specifications only weakly. The Bayes factor depends on how much prior probability $\mathcal{M}'$ assigns to small values of σ_j^2.

Remark 2 (The prior on the noise variances). Suppose the data are generated by $\mathcal{M}$. Under $\mathcal{M}'$ the posterior of σ_j^2 then concentrates near zero, and the marginal likelihood of $\mathcal{M}'$ is close to that of $\mathcal{M}$ times the prior probability of the small values of σ_j^2 that the data support. Under a fixed prior $\mathcal{IG}(\nu_\sigma/2, s_\sigma/2)$ with a small s_σ, which is the usual choice when the prior is meant to be uninformative, this probability is not small, and the two specifications are close to a tie. Under the hierarchical prior, s_σ is estimated from the n noise variances of the panel and is of the order of their typical value. The prior density of each σ_i^2 contains the factor $\exp(-s_\sigma/2\sigma_i^2)$, which is negligible when σ_i^2 is far below the typical noise variance of the panel. The specification $\mathcal{M}'$ can then approach $\mathcal{M}$ only at a cost, and the comparison favours treating an exact series as an observed factor. A series whose noise variance is comparable to those of the other series is not penalized in this way. In this sense the hierarchical prior defines an observed factor as a series whose measurement error is negligible relative to that of the rest of the panel.

The effect is large in practice. In a pilot run on a simulated panel with 100 series of which three are observed factors, the fixed prior $\mathcal{IG}(1/2, 0.005)$ on the noise variances left the true specification and two specifications that treat one of its observed factors as latent within one log point of each other. Under the hierarchical prior the two alternatives fell 14.6 and 16.5 log points below the true specification.[3]

2.6 Posterior simulation

We estimate each specification by Markov chain Monte Carlo. A Gibbs sampler draws each group of parameters in turn from its distribution given the current values of all the others. After many rounds the draws behave like draws from the posterior, and we call the sequence of draws a chain. A sampler is efficient when successive draws are nearly independent. It is slow when they are highly correlated, because many more draws are then needed for the same precision. In a large FAVAR some groups of parameters depend strongly on each other, and drawing them one group at a time is slow. This subsection shows where the problem arises and how we avoid it.

Stack $s = (s_1', \ldots, s_T')'$ and let H be the $Tr \times Tr$ block lower triangular matrix with identity diagonal blocks and with block $(t, t-p)$ equal to $-B_p$ for $1 \le p \le \min(P, t-1)$.

[3] We do not prove that the Bayes factor selects the true specification as T grows. Standard arguments (Schwarz, 1978; Kass and Raftery, 1995) imply that the log Bayes factor against a specification that cannot reproduce the distribution of the data diverges at rate T. Specifications that contain the true distribution with a redundant component, such as an additional latent factor or an exact series treated as latent, are not regular, because by Proposition 2 the true distribution lies on their boundary. The usual penalty of order $\log T$ does not apply directly to these cases, and we study them by simulation.

Then $s \sim \mathcal{N}(0, K^{-1})$ with

$$K = H'\mathcal{V}^{-1}H, \qquad \mathcal{V} = \operatorname{diag}(V_0,\ I_{T-1} \otimes \Omega)\,, \tag{5}$$

and K is block banded with bandwidth P. A banded matrix has nonzero entries only near its diagonal, which makes computations with it fast. Let K_{ff}, K_{fy} and K_{yy} denote the submatrices of K for the elements of f and y.

Lemma 1 (Prior and full conditional of the factors)**.** *Conditional on y and θ, $f \sim \mathcal{N}(\mu_f, K_{ff}^{-1})$ with $\mu_f = -K_{ff}^{-1}K_{fy}\,y$. Conditional also on Z, $f \sim \mathcal{N}(\hat{f}, Q^{-1})$ with*

$$Q = K_{ff} + (I_T \otimes \Lambda')R(I_T \otimes \Lambda), \qquad \hat{f} = Q^{-1}(K_{ff}\mu_f + (I_T \otimes \Lambda')R\tilde{z})\,.$$

In the baseline model $Q = K_{ff} + I_T \otimes G$. Both precision matrices are block banded with $r_1 \times r_1$ blocks and bandwidth P, so means, log determinants and draws require $O(TP^2r_1^3)$ operations.

Lemma 2 (Loadings on the observed factors with the factors integrated out)**.** *Let $\gamma = \operatorname{vec}(\Gamma)$ have prior $\mathcal{N}(0, V_\Gamma)$. Conditional on Z, y, Λ, Σ, B and Ω, with f integrated out, $\gamma \sim \mathcal{N}(\hat{\gamma}, D)$, where*

$$\begin{aligned} D^{-1} &= V_\Gamma^{-1} + (Y' \otimes I_n)\,W^{-1}(Y \otimes I_n), \\ \hat{\gamma} &= D\,(Y' \otimes I_n)\,W^{-1}\,(z - (I_T \otimes \Lambda)\mu_f)\,, \\ W^{-1} &= R - R(I_T \otimes \Lambda)\,Q^{-1}\,(I_T \otimes \Lambda')R. \end{aligned}$$

In the baseline model $D^{-1} = V_\Gamma^{-1} + Y'Y \otimes \Sigma^{-1} - (Y' \otimes \Sigma^{-1}\Lambda)\,Q^{-1}\,(Y \otimes \Lambda'\Sigma^{-1})$.

We draw Γ with the factors integrated out for the following reason. For any $r_1 \times r_2$ matrix C, the transformation $\mathcal{T}_C : (f_t, \Gamma, B_p, \Omega) \mapsto (f_t + Cy_t,\ \Gamma - \Lambda C,\ A_C B_p A_C^{-1},\ A_C \Omega A_C')$, with A_C the matrix A of Section 2.3 for $M_{11} = I_{r_1}$ and $M_{12} = C$, leaves both the measurement density and the VAR density unchanged. In words, part of the observed factors can be shifted into the latent factors, and the loadings Γ adjusted to compensate, without changing the fit. Under IRc this transformation leaves the parameter space unless $C = 0$, because the transformed covariance block is $\Omega_{fy} + C\Omega_{yy}$. The information that pins down $C = 0$, however, reaches the factors only through the prior precision K_{ff}. The factors and Γ are therefore strongly dependent in the posterior, and a sampler that draws them one at a time moves slowly.

Proposition 3 (Mixing of the factors and the loadings on the observed factors)**.** *In the baseline model, fix Λ, Σ, B and Ω under IRc, let the prior on Γ be flat, and let $P_Y =$*

Algorithm 1 Partially collapsed sampler for a FAVAR specification under IRc

1: If $r_2 > 0$, draw Γ from $p(\Gamma \mid \Lambda, \Sigma, B, \Omega, \tau_\Gamma^2, Z, y)$ with f integrated out (Lemma 2).
2: Draw f from $p(f \mid \Lambda, \Gamma, \Sigma, B, \Omega, Z, y)$ (Lemma 1).
3: Draw each free row Λ_i, $i > r_1$, from its Gaussian full conditional given f, Γ, σ_i^2 and τ_Λ^2.
4: Draw σ_i^2, $i = 1, \ldots, n$, from their inverse gamma full conditionals given s_σ, and under (3) update each ρ_i by the Metropolis–Hastings step of Appendix C.
5: Draw τ_Λ^2 and, if $r_2 > 0$, τ_Γ^2 from their inverse gamma full conditionals, and s_σ from $\mathcal{G}(a_\sigma + n\nu_\sigma/2,\ b_\sigma + \frac{1}{2}\sum_i \sigma_i^{-2})$.
6: Draw the coefficients of the equations for f and, if $r_2 > 0$, for y from their Gaussian full conditionals, which form two separate blocks under IRc.
7: Draw Ω_{ff} and, if $r_2 > 0$, Ω_{yy} from their inverse Wishart full conditionals.
8: Propose a joint rescaling of the factors, their free loadings and the VAR parameters by (6) and accept it with probability (7).

$Y(Y'Y)^{-1}Y'$, $g = \lambda_{\max}(G)$ *and* $\kappa = \lambda_{\max}(K_{ff})$. *The Gibbs sampler that alternates between* $f \mid \Gamma$ *and* $\Gamma \mid f$ *converges geometrically at rate*

$$\varrho^2 = \lambda_{\max}\big[(K_{ff} + I_T \otimes G)^{-1}(P_Y \otimes G)\big] \ \geq \ \frac{g}{g+\kappa},$$

which is also the lag-one autocorrelation of the slowest linear function of Γ. *The sampler that draws* Γ *from its conditional with* f *integrated out (Lemma 2) and then* $f \mid \Gamma$ *produces independent draws of* (f, Γ).

Since $g = \lambda_{\max}(\Lambda'\Sigma^{-1}\Lambda)$ grows linearly with the number of series that load on the factors, the rate of the standard sampler approaches one exactly when the factors are well measured, and in particular as the panel grows. A rate close to one means that each draw is almost a copy of the previous one. In the simulation design of Appendix A.1 with the weak observed factor, evaluated at the least squares estimates obtained with the true factor, the rate is 0.991 with 100 series and 0.996 with 250 series. These rates imply that about 220 and 450 draws are needed to obtain the equivalent of one independent draw of the slowest linear function of Γ. Algorithm 1 therefore draws Γ and f as one block. The last statement of Proposition 3 holds under (3) as well, since it uses only the joint normality of (f, Γ) given the other parameters.

Proposition 4 (Validity)**.** *Algorithm 1 leaves the joint posterior* $\pi(f, \theta, \tau_\Lambda^2, \tau_\Gamma^2, s_\sigma \mid Z, y)$ *invariant, in the baseline model and under* (3). *When* $r_2 = 0$ *the same holds without* Γ *and* τ_Γ^2.

Steps 1 and 2 must be taken in this order and without an update of f in between, because together they form one draw from the joint conditional distribution of (Γ, f) (van Dyk and Park, 2008).

Step 8 addresses a second slow direction, which appears in large panels. Under IRc the scale of each latent factor is fixed by its anchor series alone. Multiplying f_{kt} by c_k for all t and dividing the free loadings in column k of Λ by c_k leaves the common component of every other series unchanged. Along this path the posterior is shaped only by the anchor equation, the VAR and the priors. Steps 2 and 3 update the factors and the loadings separately, so they move along the path slowly when many series load on the factor. Step 8 moves along it directly. Draw $\log c_k \sim \mathcal{N}(0, \delta^2)$ independently for $k = 1, \dots, r_1$, let $C = \operatorname{diag}(c_1, \dots, c_{r_1})$ and $A_c = \operatorname{diag}(C, I_{r_2})$, and propose

$$f_t \mapsto C f_t, \qquad \Lambda_2 \mapsto \Lambda_2 C^{-1}, \qquad B_p \mapsto A_c B_p A_c^{-1}, \qquad \Omega_{ff} \mapsto C\,\Omega_{ff}\,C, \tag{6}$$

with $\Lambda_1 = I_{r_1}$, Γ, Σ, Ω_{yy} and the scales unchanged. The proposal satisfies IRc and is accepted with probability

$$\min\left\{1,\ \frac{\pi(f^c, \theta^c, \tau_\Lambda^2, \tau_\Gamma^2 \mid Z, y)}{\pi(f, \theta, \tau_\Lambda^2, \tau_\Gamma^2 \mid Z, y)} \prod_{k=1}^{r_1} c_k^{T-(n-r_1)+r_1+1}\right\}, \tag{7}$$

where (f^c, θ^c) is the proposed state and the product is the Jacobian of (6). It counts T factor values and $n - r_1$ free loadings for each factor, and the factor $c_k^{r_1+1}$ comes from Ω_{ff}. The map $B_p \mapsto A_c B_p A_c^{-1}$ has unit Jacobian. Only the anchor equations, the VAR density, the priors on Λ_2, B and Ω_{ff} and the Jacobian change, so the ratio is cheap to evaluate. The move is a Metropolis–Hastings version of the generalized Gibbs step of Liu and Sabatti (2000) and is related to the parameter expansion of Liu and Wu (1999). The step size δ is adapted during the burn-in, the initial draws that are discarded, to an acceptance rate of about 0.4, and then held fixed.

Under (3) each step keeps its form after the regressions of the panel series are transformed with ρ_i, and ρ_i is updated by an independence Metropolis–Hastings step. Appendix C gives the details. A specification then takes about twice as long to estimate as in the baseline model.

2.7 Marginal likelihood

The marginal likelihood of a FAVAR is an integral over the factors, the loadings and the remaining parameters, and the factors alone contribute Tr_1 dimensions. Three features make it tractable here. First, the factors and Γ are integrated out exactly (Lemma 3). Simulation is then needed only for parameters whose number does not grow with T and whose posterior is close to Gaussian once variances and covariance matrices are mapped to unconstrained

coordinates. Second, Algorithm 1 produces draws of these parameters that mix well in large panels (Proposition 3 and Step 8). Third, the specifications are estimated independently, so the specifications visited in each step of the search of Section 2.8 can be estimated in parallel.

Let $\phi = (\Lambda_2, \Sigma, B, \Omega_{yy}, \Omega_{ff}, \tau_\Lambda^2, \tau_\Gamma^2)$ denote the parameters that remain after the factors and Γ are integrated out. The scale s_σ is not part of ϕ, because the prior of Σ in $\pi(\phi)$ is (4), in which it is integrated out. When $r_2 = 0$, ϕ contains neither Ω_{yy} nor τ_Γ^2, and under (3) it also contains ρ. The marginal likelihood is $m(X \mid \mathcal{M}) = \int p(Z, y \mid \phi)\, \pi(\phi)\, d\phi$, where $p(Z, y \mid \phi)$ is available in closed form.

Lemma 3 (Integrated likelihood). *(a) With Γ fixed, $\log p(Z, y \mid \theta) = \log p(Z \mid y, \theta) + \log p(y \mid \theta)$, where*

$$\begin{aligned}
\log p(Z \mid y, \theta) = &-\frac{Tn}{2}\log(2\pi) + \frac{1}{2}\log|R| - \frac{1}{2}\big(\tilde{z} - (I_T \otimes \Lambda)\hat{f}\big)' R \big(\tilde{z} - (I_T \otimes \Lambda)\hat{f}\big) \\
&- \frac{1}{2}(\hat{f} - \mu_f)' K_{ff} (\hat{f} - \mu_f) + \frac{1}{2}\log|K_{ff}| - \frac{1}{2}\log|Q|, \\
\log p(y \mid \theta) = &-\frac{Tr_2}{2}\log(2\pi) + \frac{1}{2}\log|K| - \frac{1}{2}\log|K_{ff}| - \frac{1}{2}\, y' \big(K_{yy} - K_{yf} K_{ff}^{-1} K_{fy}\big)\, y.
\end{aligned}$$

(b) With $\gamma \sim \mathcal{N}(0, \tau_\Gamma^2 I)$ integrated out, for any value γ^*,

$$\log p(Z, y \mid \phi) = \log p(Z, y \mid \phi, \gamma^*) + \log \mathcal{N}(\gamma^* \mid 0, \tau_\Gamma^2 I) - \log \mathcal{N}(\gamma^* \mid \hat{\gamma}, D),$$

with $\hat{\gamma}$ and D from Lemma 2. We evaluate it at $\gamma^* = \hat{\gamma}$. When $r_2 = 0$ there is no γ and $p(Z, y \mid \phi) = p(Z \mid \theta)$ is given by (a).

In the baseline model the first terms of $\log p(Z \mid y, \theta)$ reduce to $-\frac{T}{2}\log|\Sigma| - \frac{1}{2}\sum_t (\tilde{Z}_t - \Lambda \hat{f}_t)' \Sigma^{-1} (\tilde{Z}_t - \Lambda \hat{f}_t)$.[4]

The remaining integral over ϕ has no closed form, and we estimate it by bridge sampling (Meng and Wong, 1996). The idea is to compare the posterior draws with draws from a simple distribution whose integral is known, here a Gaussian, and to use the two sets of draws to estimate the ratio of the two integrals. The draws of ϕ from Algorithm 1 are first mapped to unconstrained coordinates, by taking logarithms of the variances and scales, Cholesky factors with log diagonals of Ω_{yy} and Ω_{ff} and, under (3), $\operatorname{atanh}\rho_i$, with the corresponding Jacobians added to the log prior. The mean and covariance of the Gaussian proposal are estimated from one half of the draws, and the other half enters the bridge identity (Overstall

[4] Under (3), with all $\rho_i = 0$ our implementation reproduces the log posterior of the baseline model to within 10^{-13}. With nonzero ρ_i the integrated likelihood of Lemma 3 agrees to within 10^{-12} with a Kalman filter that carries the errors in its state.

and Forster, 2010). The target density is evaluated exactly through Lemma 3, so the only simulation error comes from the bridge estimator. As a cheaper alternative we also compute the Laplace–Metropolis estimator of Lewis and Raftery (1997), which replaces the posterior of ϕ by a Gaussian with the mean and covariance of the draws. Integrating out f and Γ matters for both estimators. The posterior of the remaining parameters is much closer to Gaussian than the joint posterior, and its dimension does not grow with T or with the number of loadings on the observed factors.

Both estimators rely on a Gaussian proposal or approximation whose mean and covariance are estimated from the draws, and the dimension of ϕ grows with Nr_1. With several latent factors it exceeds 700 in a panel of 100 series, and in this range the number of draws matters. For a specification with six latent factors and 100 series, bridge sampling from 4000 retained draws overstated the log marginal likelihood by about 50 log points, while with 16000 retained draws two chains agreed to within 0.1 log points.[5] Too few draws therefore favour specifications with many latent factors, which is the direction that matters for the joint choice of S and r_1. We use 1000 burn-in and 16000 retained draws, and we report the difference between the bridge and Laplace–Metropolis estimates for every specification as a diagnostic. Section 3.2 reports how the two estimates compare in the simulations. One specification with 100 series takes about ten minutes on one core of a laptop with these settings.

2.8 Searching over specifications

When r_1 is free and any series may enter the VAR, there are far too many specifications to estimate them all. With 100 series, the number of possible sets of observed factors alone exceeds 10^{30}. We therefore restrict the candidates for the VAR to a short list and search over the specifications built from it. Algorithm 2 summarizes the procedure.

The short list. An observed factor is a state of the VAR, so it is an exact linear combination of the $r = r_1 + r_2$ factors of the panel. A series that loads on the factors with noise is not. Principal components estimate the space spanned by the factors, so an observed factor should be explained almost completely by them. We estimate r by the IC_{p2} criterion of Bai and Ng (2002), extract $k = \hat{r} + 2$ principal components from the standardized panel, and compute for each series the share $u_j = 1 - R_j^2$ of its variance that the components do

[5] With 4000 draws, estimates from independent chains differed by up to 11 log points. With 16000 draws they agreed with the Laplace–Metropolis estimate to within one log point. For the true specification, with three latent factors, the estimates agreed to within one log point at both chain lengths. These comparisons used the fixed prior on the noise variances.

not explain. The short list $\mathcal{C}$ consists of the J series with the smallest u_j, together with any series that the researcher wants to consider on economic grounds. The two components beyond $\hat{r}$ guard against underestimating r. In our simulations IC_{p2} estimated r correctly except when an observed factor had weak dynamics, and then it estimated one factor too few. Let $u_j^* = \mathrm{Var}(\varepsilon_{jt})/\mathrm{Var}(X_{jt})$ denote the population noise share of series j, so that $u_j^* = 0$ for an observed factor, and let $\hat{u}_j(k)$ denote u_j computed with k components.

Proposition 5 (The short list)**.** *Let S_0 with $|S_0| = r_2$ be the true set of observed factors, and suppose that as $N, T \to \infty$ (i) $\Pr(\hat{r} \geq r - 2) \to 1$, (ii) for every fixed $k \geq r$, $\max_j |\hat{u}_j(k) - u_j^*| \to 0$ in probability, where the maximum is over all series, and (iii) $u_j^* \geq c$ for some $c > 0$ and all $j \notin S_0$. Then $\Pr(S_0 \subseteq \mathcal{C}) \to 1$ for every $J \geq r_2$.*

Condition (i) holds when IC_{p2} is consistent, which Bai and Ng (2002) establish under their assumptions, and it allows $\hat{r}$ to fall short of r by up to two. Condition (ii) requires the fitted common components to be uniformly close to the true ones. For $k = r$ results of this kind are part of the theory of principal components in large factor models (Bai, 2003; Bai and Ng, 2006). For $k > r$ the condition also requires that the additional components absorb a vanishing share of the variance of every series, which we expect when the idiosyncratic errors are weakly dependent across series but do not prove here. Condition (iii) rules out series that are nearly exact factors without being states of the VAR. Such series compete with the observed factors for a place on the short list, and the marginal likelihood must separate them from the observed factors. The near-exact design of Section 3 shows that this works in most data sets but not in all.

The search. The specifications are $\mathcal{M} = (S, r_1)$ with $S \subseteq \mathcal{C}$, $1 \leq r_1 \leq r_{\max}$ and the lag order fixed. From (S, r_1) the search can move to six kinds of neighbouring specifications. It can add a series of $\mathcal{C}$ to S, remove a series from S, raise or lower r_1 by one, promote a series, which adds it to S and lowers r_1 by one, or demote a series, which removes it from S and raises r_1 by one. Promotion and demotion keep r fixed and carry out the comparison of Proposition 2 in one step. The search starts from S equal to the series that the researcher keeps in the VAR, which may be none, and $r_1 = \max(1, \hat{r} - |S|)$, so that the starting specification has the estimated number of factors. Each step estimates all neighbours of the best specification whose neighbours have not yet been estimated, which we call expanding it. The search stops when every specification within $\log W$ of the best and each of the three best specifications have been expanded, and we use $W = 100$. The search thus keeps exploring around every specification whose posterior probability is at least one hundredth of that of the best. The second condition matters when the path to the best specification passes through

Algorithm 2 Search over specifications

1: Standardize X, compute $\hat{r}$ by IC_{p2} and $\hat{u}_j$ from $\hat{r}+2$ principal components, and form the short list $\mathcal{C}$ from the J series with the smallest $\hat{u}_j$ and any series added on economic grounds.
2: Set S to the series kept in the VAR and $r_1 = \max(1, \hat{r} - |S|)$, and estimate $\log m(X \mid S, r_1)$ by Algorithm 1 and bridge sampling, using the two chain lengths described in the text.
3: **repeat**
4: Among the specifications estimated so far and not yet expanded, take the one with the largest marginal likelihood among those within $\log W$ of the best or among the three best.
5: Estimate every adjacent specification not estimated before, in parallel, and mark the specification as expanded.
6: **until** no estimated specification within $\log W$ of the best or among the three best remains unexpanded
7: Report $\Pr(\mathcal{M} \mid X)$ over the estimated specifications, $\Pr(j \in S \mid X)$ for $j \in \mathcal{C}$, $\Pr(r_1 \mid X)$ and responses averaged over specifications.

a specification that is worse than the current best by more than $\log W$.[6] Each specification is estimated only once at each chain length, and the results are stored.

Two chain lengths. Most of the specifications a search visits lie far below the best, and their marginal likelihoods need not be estimated precisely. Each new specification is therefore first estimated with a short chain of 500 burn-in and 4000 retained draws. Every specification whose estimate from this short chain is within $\Delta = 30$ log points of the current best is estimated again with a long chain of 1000 burn-in and 16000 retained draws, and the estimate from the long chain replaces the short one. The step is repeated until no specification within Δ of the best lacks a long chain. In our experience short chains overstate the marginal likelihood of large specifications (Section 2.7), so a specification that is more than Δ below the best on a short chain is unlikely to become the best on a long one.

Reporting. Posterior probabilities are computed over the estimated specifications under equal prior probabilities. Specifications far below the best carry negligible probability, so restricting the average to the specifications near the best changes little. This is the argument for the Occam's window of Madigan and Raftery (1994). From these probabilities we report the inclusion probability of each series on the short list, the posterior distribution of r_1, and responses averaged over specifications. Like any local search, the procedure can miss a specification that is better than all specifications near its path. The comparisons in which

[6] In a data set of the weak-factor design of Section 3, the route to the true specification led through a specification 9.5 log points below the best, and without the condition the search stopped before reaching it.

the marginal likelihood changes least, those of Proposition 2, are single moves, which reduces this risk, and the simulations of Section 3 assess it.

An illustration. In the first data set of the baseline design of Section 3, with 100 series, three observed factors and three latent factors, IC_{p2} gave $\hat{r} = 6$, and the three observed factors left at most 3.2 percent of their variance unexplained by the principal components, against at least 14.6 percent for every other series. Starting from six latent factors and no observed factors, the search moved one series at a time into the VAR, first the policy rate, then x_1 and then x_2, gaining 22, 16 and 27 log points. It stopped after 74 specifications. The closest alternatives treated the policy rate or x_2 as a latent factor and were 23.1 and 26.9 log points lower.

3 Simulation evidence

3.1 Design

We first test the procedure on simulated data, where the true specification is known. The simulated panels have the size of those used in applied work. Bernanke et al. (2005) use 120 monthly series and Boivin et al. (2009) more than six hundred, and the FRED-MD and FRED-QD databases (McCracken and Ng, 2016, 2021) contain about 130 and 250 series. We simulate panels of $N = 100$ series with $T = 200$.

The true VAR contains three observed factors, the policy rate and two further series x_1 and x_2, and three latent factors. It is a VAR(1) with own-lag coefficients 0.7, 0.6 and 0.65 for the observed factors and 0.6 for the latent factors, and cross effects drawn from $U(-0.15, 0.15)$. The coefficient matrix is scaled down until its spectral radius is at most 0.9, which keeps the VAR stationary. The innovations of the observed factors have unit variances and correlations 0.3, 0.2 and -0.2. The innovations of the latent factors are uncorrelated with them and with each other, so that the true parameters satisfy IRc. Three further series x_3, x_4 and x_5 are decoys, series built to look like observed factors without being states of the VAR. Each is a linear combination of the observed factors, with coefficients drawn from $U(-0.5, 0.5)$, and of the latent factors, with coefficients drawn from $U(0.3, 1)$, plus noise whose variance is a share σ_d^2 of the variance of this signal. The other 94 series load on the observed factors with coefficients drawn from $U(-0.8, 0.8)$ and on the latent factors with coefficients drawn from $U(-1, 1)$, and their noise shares are drawn from $U(0.2, 0.8)$. The first three of them each load on one latent factor only, with noise share 0.3. Many series in macroeconomic panels are only loosely related to the common factors. To reflect this,

each of the remaining series is, with probability one third, a weak series whose loadings are multiplied by 0.2. All series are standardized.

Nothing about the specification is given to the procedure. No series is kept in the VAR, so the data decide whether the policy rate is an observed factor, and r_1 ranges from one to $r_{\max} = 8$. The search of Section 2.8 forms a short list of $J = 8$ series, starts from $\hat{r}$ latent factors and no observed factors, and runs until its stopping rule is met. A specification may contain up to six observed factors. We use the hierarchical priors of Section 2.5 for the loadings and for the noise variances, with $a_\tau = b_\tau = 1/2$, $\nu_\sigma = 1$, $a_\sigma = b_\sigma = 1$, $b = 0.25$ and $V_0 = 10I$. The simulated errors are serially uncorrelated, and we estimate the baseline model.

There are three designs, with 20 data sets each. In the baseline design $\sigma_d^2 = 0.3$. In the near-exact design $\sigma_d^2 = 0.01$, so that the decoys are nearly exact linear combinations of the factors and can stand in for an observed factor or for a latent one. The weak-factor design has $\sigma_d^2 = 0.3$, but the effect of x_2 on every factor, including its own lag, and all loadings on x_2 are multiplied by 0.3. Over the 20 data sets the median first-order autocorrelation of x_2 is then 0.28, against 0.68 without the scaling, and the lagged factors explain about seven percent of its variance. The baseline design uses seeds 601 to 620, and the near-exact and weak-factor designs share seeds 801 to 820. None of these seeds was used while the method was developed.

3.2 Results

Table 1 reports, for each design, the number of data sets in which the search selects the true specification $\mathcal{M}_0$, with the true observed factors and $r_1 = 3$. It also reports the average posterior probability of $\mathcal{M}_0$, the median and smallest margin between $\mathcal{M}_0$ and the best other specification visited, the average posterior probabilities that $r_1 = 3$, that x_2 is in the VAR and that a decoy is in the VAR, and the number of specifications the search estimated. All posterior probabilities are computed over the visited specifications under equal prior probabilities.

Screening and search. In all 60 data sets the short list contained the three observed factors, and the search visited the true specification. In two data sets of the weak-factor design the true specification was more than Δ below the best on its short chain and was not re-estimated with a long one. Both are among the errors discussed below, with the truth more than 40 log points behind. IC_{p2} gave $\hat{r} = 6$ in every data set of the baseline and near-exact designs, so the search started from a pure factor model with six latent factors. In

Table 1: Joint choice of the observed factors and the number of latent factors, $N = 100$

Design	Correct	$\Pr(\mathcal{M}_0 \mid X)$	Margin		$\Pr(r_1 = 3)$	$\Pr(x_2 \in S)$	$\Pr(\text{decoy} \in S)$	Visited
			Median	Min				
Baseline	20/20	1.000	22.9	10.1	1.000	1.000	0.000	73
Near-exact	19/20	0.950	18.2	−13.0	0.950	1.000	0.050	74
Weak x_2	11/20	0.517	1.3	−46.7	0.628	0.577	0.000	72

Notes. Three observed and three latent factors, $T = 200$, 20 data sets per row, no series kept in the VAR. Correct is the number of data sets in which the true specification $\mathcal{M}_0$ has the highest estimated marginal likelihood among the visited specifications. Margin is $\log m(X \mid \mathcal{M}_0)$ minus the largest log marginal likelihood among the other visited specifications, so negative values mean that another specification is preferred. $\Pr(\text{decoy} \in S)$ is the largest average inclusion probability among x_3, x_4, x_5. Visited is the median number of specifications estimated by the search. Hierarchical priors on the loadings and on the noise variances.

the weak-factor design it gave $\hat{r} = 5$ in every data set, one factor too few, so the search had to raise the total number of factors as well as move series into the VAR.

Baseline. The true specification is selected in all 20 data sets, with posterior probability above 0.999 and margins between 10.1 and 31.6 log points. In 18 data sets the closest competitor treats one of the observed factors as an additional latent factor, which is the comparison of Proposition 2. In the other two it adds a fourth latent factor to the true specification. The hierarchical prior on the noise variances therefore separates an observed factor from a latent factor anchored on the same series by at least ten log points in every data set. No decoy entered a specification with appreciable posterior probability.

Near-exact decoys. The true specification is selected in 19 data sets, with margins between 4.8 and 26.1 log points. In four of them the closest competitor adds a decoy to the VAR and drops one latent factor, and the smallest margin, 4.8 log points, is of this kind. In the remaining data set the search selects the policy rate, x_1, x_2 and the decoy x_5 with two latent factors, 13.0 log points above the true specification. The decoy carries only one percent noise, so the observed factors, the decoy and two latent factors span nearly the same space as the truth, with about 95 fewer parameters. The two models imply nearly the same distribution of the data, and the bridge and Laplace–Metropolis estimates agree to within one log point for both, so the ranking is not a numerical artefact. When a series is almost an exact factor, the marginal likelihood can prefer it in place of a latent factor. This is a property of the data rather than a failure of the search.

A weak observed factor. The true specification is selected in 11 of 20 data sets. When it is selected the margins are smaller than in the other designs, between 0.5 and 19.0 log points

with a median of 5.0. The posterior probabilities of the true specification reflect this and fall to 0.63 and 0.86 in the two closest cases. In six of the nine other data sets the selected specification treats x_2 as a noisy indicator of an additional latent factor, in two it leaves x_2 in the panel, and in one it keeps x_2 but drops the policy rate and x_1.[7] These outcomes follow from the design. With its dynamics and its loadings scaled by 0.3, x_2 contributes little that an additional latent factor anchored on it, or the panel itself, cannot reproduce. This is the situation that Proposition 2 and Remark 2 describe. The procedure misses a weak observed factor in about half the data sets, most often by treating it as latent, and it never adds a decoy.

Computation. In the weak-factor design the bridge and Laplace–Metropolis estimates differ by a median of 13 and up to 32 log points for the specifications that replace x_2 by a latent factor, the boundary specifications of Proposition 2, with the Laplace–Metropolis estimate always higher. Selection by it would change the outcome in seven of the 20 data sets, in five of them from the true specification to the one that treats x_2 as latent. In two of these five we re-estimated both specifications with 64000 retained draws. The bridge estimates changed by at most half a log point with the length of the chain and across chains and proposal covariances, while the Laplace–Metropolis estimates kept rising and moved further from them.[8] We read this as a failure of the Gaussian approximation near the boundary and regard the bridge estimates as reliable. Elsewhere the two estimators differ by less than one log point at the median for specifications within ten log points of the best, over 929 long-chain estimates.

Each search estimated between 55 and 83 specifications with short chains and re-estimated a median of 15 with long chains. A short chain takes about three minutes and a long chain about ten minutes on one core of a laptop, and the 60 searches together used about 380 core hours. Step 8 of Algorithm 1 lowers the inefficiency factor of the factor scale, the number of draws needed for the equivalent of one independent draw, from 74 to 8.8 in a pilot data set, and the median for the free loadings from 15 to 3.2.

[7] The truth lies between 7.2 and 8.7 log points below the selected specification in four of the nine data sets, and between 23.2 and 46.7 below in the other five.

[8] Between 16000 and 64000 draws the Laplace–Metropolis estimate rose by 2.2 and 2.4 log points for the true specification and by 3 to 4 log points for the latent alternative, and its distance from the bridge estimate grew to 20 and 15 log points. Two independent chains of the latent alternative agreed to within 0.1 log points, and the bridge estimate stayed within one log point when the Gaussian proposal used a shrunk or a block-diagonal covariance. In one of the two data sets the two specifications are within 0.1 log points of each other.

Further evidence. Appendix A reports two further sets of simulations. The first fixes the number of latent factors and compares the hierarchical prior on the loadings with a fixed-scale prior. The fixed-scale prior drops a weak observed factor in every data set with 100 series, by up to 274 log points, while the hierarchical prior keeps it in most data sets. The second changes the hyperpriors one at a time. The parameter ν_σ matters most. A stronger pull of the noise variances toward their common scale widens the margins of genuine observed factors but also lets nearly exact decoys into the VAR. The hyperprior on the loading scales matters only when the observed factors of a specification all have small loadings.

4 Application

We now return to the question with which the paper began, the choice of observed factors in a FAVAR for monetary policy. The application differs from the simulations in two ways that turn out to matter. The idiosyncratic errors of a macroeconomic panel are serially correlated, and the number of latent factors that the marginal likelihood favours is larger than the number that can be estimated reliably with our chain lengths.

4.1 Data and short list

The panel is built from the September 2026 vintage of FRED-QD (McCracken and Ng, 2021), a public database of quarterly US macroeconomic series. We use 1960Q1 to 2007Q4, so that $T = 192$ and the sample ends before the federal funds rate reached its lower bound. The panel contains the series that McCracken and Ng (2021) use to estimate factors and that are available over the whole sample, together with four aggregates that applied work treats as observed factors, namely industrial production, the consumer price index, the unemployment rate and real M2. So that no candidate is an accounting combination of other series in the panel, we drop the seven components of industrial production and the three unemployment rates by demographic group. This leaves $N = 95$ series. We apply the transformations of McCracken and Ng (2021), such as log differences for series that trend, with two exceptions that follow Bernanke et al. (2005). Price indices enter as inflation rates rather than as changes in inflation, and the federal funds rate enters in levels. No observation lies more than ten interquartile ranges from the median of its series, the outlier rule of McCracken and Ng (2016), so none was replaced. All series are standardized.

The IC_{p2} criterion gives $\hat{r} = 5$. With seven principal components, the eight series with the smallest unexplained shares u_j are CPI inflation (0.081), inflation in the price deflator of the business sector (0.084), employment in trade, transportation and utilities (0.120),

employment in durable goods manufacturing (0.134), real household assets excluding real estate (0.136), the unemployment rate (0.151), real household net worth (0.156) and capacity utilization in manufacturing (0.165). The median share over the panel is 0.43. We add the federal funds rate, industrial production, real M2 and inflation in the producer price index for industrial commodities on economic grounds, which gives a short list of twelve series. The federal funds rate is kept in the VAR in every specification, so that the policy innovation is defined in all of them, and the comparison concerns the other eleven. We use one lag and the priors and chain lengths of Section 3.

4.2 Serially correlated errors and the number of latent factors

A first search with the baseline model of Section 2 did not settle. With r_1 free, it reached the largest number of latent factors allowed, $r_{\max} = 8$, and had not met its stopping rule after 128 specifications. The data show why. After the first five principal components are removed, the residuals of the panel series have a median first-order autocorrelation of 0.18, and 22 percent of them have an autocorrelation above 0.5. In the simulated panels of Section 3 the median is -0.02. A model with serially uncorrelated errors can account for this persistence only through the factors. It therefore adds latent factors, and its leading specifications put smooth balance sheet series in the VAR.

We therefore let each idiosyncratic error follow its own first-order autoregression (3), with the prior on ρ_i and the computations of Section 2. The extension costs little when it is not needed. In the illustration of Section 2.8, whose errors are serially uncorrelated, it leaves the margins of the true specification over its two closest competitors at 23.8 and 27.3 log points, against 23.1 and 26.9 without it. On the application panel the log marginal likelihood rises by between 1641 and 2081 log points in all 13 specifications that we estimated under both models with short chains. In the specification selected below, 23 percent of the posterior means of the ρ_i exceed 0.5, and the largest belong to the term spreads, the credit spread and average weekly hours in manufacturing. We use the model with serially correlated errors in the rest of this section.

The number of latent factors. With serially correlated errors and r_1 free, the search again moved to $r_1 = 8$. The best specifications with 4, 5, 6, 7 and 8 latent factors gained 146, 178, 297 and 226 log points over one another. Re-estimating the best specification with 8 latent factors with 9, 10 and 12 latent factors raised its log marginal likelihood by 112, 197 and 405 log points relative to eight. On this panel the marginal likelihood keeps rewarding additional latent factors at least up to twelve, far beyond the $\hat{r} = 5$ of IC_{p2}.

These comparisons are not reliable. Three further long chains of the best specification with eight latent factors, which has 1038 parameters in ϕ, gave bridge estimates that, together with the estimate in the search, span 63 log points, and in two of them the log posterior was still rising late in the run. Short chains of the same specification lay 150 to 190 log points above the long ones.[9] With five and six latent factors the problem disappears. For two specifications at each of these values, three long chains agreed to within 1.9 log points, and the log posterior was stable along every chain. We conclude that with $T = 192$ and our chain lengths the marginal likelihood cannot be used to choose r_1 on this panel. We fix r_1 at $\hat{r} = 5$ and report $r_1 = 4$ and $r_1 = 6$ as checks.

4.3 The observed factors

For each $r_1 \in \{4, 5, 6\}$ we ran the search of Section 2.8 with r_1 held fixed and the federal funds rate in the VAR, so that the search only adds and removes series of the short list. A specification may contain all twelve series, so no limit on the size of the VAR binds. Table 2 reports the results, together with the two specifications of Bernanke et al. (2005).

With five latent factors the search selects a VAR of eight series. They are the federal funds rate, industrial production, capacity utilization, the unemployment rate, employment in trade, transportation and utilities, CPI inflation, PPI inflation and real household assets excluding real estate. The second best specification drops the unemployment rate and lies 24.5 log points lower, so the selected specification has posterior probability above 0.9999 among those visited. Every specification with nine or ten series is at least 30 log points lower in this search. With a common anchor set, Section 4.4 finds several nine and ten series specifications within about 15 log points, and the eight series specification is still preferred. With four latent factors the search selects the same eight series, and the second best specification adds employment in durable goods manufacturing and lies 11.4 log points lower. Three further long chains of each selected specification gave estimates within 3.3 log points of one another and of the value in the search. The bridge estimates lie 14 to 19 log points below the Laplace–Metropolis estimates, the same direction as in the simulations.

The selection is stable across the three values of r_1. With six latent factors the search had not met its stopping rule after 120 specifications. Its best specification keeps six of the seven series, replaces household assets by the closely related household net worth, and adds employment in durable goods manufacturing and business sector inflation. Three further

[9] The bridge estimates were -17614.1, -17647.3 and -17636.7, against -17584.0 in the search, and the short chains gave -17452.3 and -17442.7. The upward bias of short chains grows with r_1, from 34 to 38 log points with five latent factors to 70 to 77 with six, so the first stage of the search favours specifications with many latent factors and adds to the problem.

Table 2: Observed factors in a monetary FAVAR, FRED-QD 1960Q1–2007Q4

	$r_1 = 4$	$r_1 = 5$	$r_1 = 6$
Federal funds rate only	−19042.4	−18474.7	
Industrial production, CPI, federal funds rate	−18964.3	−18415.4	
Selected specification	−17951.3	−17831.3	−17539.9
Margin over the second best	11.4	24.5	
Observed factors in the selected specification	8	8	10
Specifications estimated	93	102	120
Re-estimated with long chains	21	30	40
Range over three further chains	0.7	3.3	2.2

Notes. Log marginal likelihoods estimated by bridge sampling with 1000 burn-in and 16000 retained draws, serially correlated idiosyncratic errors, the federal funds rate in the VAR in every specification and r_1 fixed. With $r_1 = 4$ and $r_1 = 5$ the selected specification contains the federal funds rate, industrial production, capacity utilization in manufacturing, the unemployment rate, employment in trade, transportation and utilities, CPI inflation, PPI inflation for industrial commodities and real household assets excluding real estate. With $r_1 = 6$ the search had not met its stopping rule, and the reported specification is the best one visited. It contains the same series except household assets, together with real household net worth, employment in durable goods manufacturing and inflation in the business sector price deflator. The last row is the range of the bridge estimates of the selected specification over three further long chains from different seeds.

chains of this specification, which has 978 parameters in ϕ, agreed to within 2.2 log points, so the reliability problem above comes from the number of latent factors rather than from the number of parameters alone.

The selected VAR contains the three series of Bernanke et al. (2005) and adds five more. Their specification with industrial production, consumer prices and the federal funds rate is preferred to the federal funds rate alone, by 78 log points with four latent factors and 59 with five, so the marginal likelihood agrees that these series belong in the VAR. It does not agree that they are enough. The selected specification is 1013 and 584 log points above theirs. All seven series that the search adds to the federal funds rate have unexplained shares between 0.08 and 0.21 in the principal components step, against a median of 0.43 over the panel. The short list and the marginal likelihood therefore agree on which series are best measured by the factors. None of them is close to an exact factor. Industrial production, for example, leaves about a fifth of its variance unexplained by seven components, consistent with the finding of Bai and Ng (2006) that it is not an exact factor. With r_1 fixed, the alternative to placing such a series in the VAR is to leave it in the panel, the comparison of Appendix A.1, and each of the seven series improves the fit by more than the cost of the loadings that move with it. The least conventional series in the selected VAR is household

assets excluding real estate, which consist mostly of financial assets. Its idiosyncratic error has its own autoregression in every specification, so its persistence alone does not explain its place in the VAR. With six latent factors it gives way to household net worth, which suggests that the VAR needs a measure of household wealth rather than this particular series.

4.4 Sensitivity to the prior and to the anchors

Appendix A.2 found that ν_σ is the hyperparameter that matters most, so we re-estimated seven specifications with five latent factors under $\nu_\sigma = 4$. They are the selected specification, the four closest alternatives visited by the search, and the two specifications of Bernanke et al. (2005). The conclusions about the conventional specifications do not change. Industrial production, consumer prices and the federal funds rate are preferred to the federal funds rate alone by 69 log points, and the selected specification is 549 log points above them. At the top, however, the ranking changes. The specification that adds employment in durable goods manufacturing to the selected one lies 34 log points below it under the default prior and 13 above it under $\nu_\sigma = 4$. Two further chains of each under $\nu_\sigma = 4$ reproduced both values to within 2.1 log points, so the reversal is not simulation error.

The reversal comes from the anchors rather than from the added series. Recall from Section 2.3 that each latent factor is measured in the units of an anchor series, and that the rule there chooses the anchors separately in each specification. The two specifications end up with different anchors. Those of the selected specification include household financial assets, a panel series distinct from the household assets in the VAR, whose noise variance is about 0.006 in fits with these anchors and about 0.05 with the other set. A stronger pull toward the common scale penalizes such a small noise variance, as in Remark 2, so these specifications lose 54 to 58 log points under $\nu_\sigma = 4$, against 10 to 33 for the others. Estimating each specification with the other's anchors separates the two effects. With the anchors held fixed, the selected specification is preferred in all four comparisons, by 8 to 15 log points, while exchanging the anchors moves each log marginal likelihood by 22 to 28 log points. The second best specification of Table 2, which drops the unemployment rate, has the same anchors as the selected one, so the margin of 24.5 log points reported there is not affected.

These results suggest removing the anchors from the comparison altogether, by choosing one anchor set for all specifications and repeating the search with it. We chose the set by marginal likelihood in the selected specification, replacing each anchor of the rule of Section 2.3 in turn by the series with the second or third largest varimax loading on the

Table 3: Log marginal likelihoods with a common anchor set, $r_1 = 5$

Specification	Chains			Mean
Selected	−17815.6	−17814.4	−17812.0	−17814.0
Selected + BS inflation	−17807.5	−17828.0	−17832.7	−17822.7
Selected + BS inflation + durables − unemployment	−17818.5	−17826.9	−17825.2	−17823.5
Selected + BS inflation + durables	−17814.7	−17841.4	−17829.5	−17828.5
Selected + durables	−17830.0			
Selected − unemployment	−17832.4			
IP, CPI, FFR	−18322.9			
FFR only	−18492.7			

Notes. Bridge sampling estimates with 1000 burn-in and 16000 retained draws, serially correlated idiosyncratic errors and the default prior. Every specification uses the same anchors, chosen by marginal likelihood in the selected specification as described in the text. Each chain uses a different seed. BS inflation is inflation in the business sector price deflator, and durables is employment in durable goods manufacturing. IP, CPI, FFR is the specification of Bernanke et al. (2005) with industrial production, consumer prices and the federal funds rate.

same component.[10] We then repeated the search of Section 4.3 with this set fixed in every specification. It visited 102 specifications, and its leading specifications are built on the same series as the selected one. Four specifications ended within 11 log points of one another, the selected one and three that add inflation in the business sector price deflator, two of them also with employment in durable goods manufacturing. The selected specification already contains CPI and PPI inflation, so these near ties add a third inflation series to it. In the search the specification that adds only business sector inflation was ranked highest, but on a single chain. Table 3 reports three chains for each of the four.

The selected specification has the highest mean, 8.7 log points above the next, and its three estimates lie within 3.6 log points of one another. The estimates for the three specifications with business sector inflation vary by 8 to 27 log points across chains, and each of them was ranked near the top in the search because of its highest chain. With a common anchor set the selected specification is therefore still preferred, but by about 9 log points rather than the 24.5 of Table 2. The large margins are unaffected. The selected specification is 509 log points above the conventional one, and the conventional specification is 170 log points above the federal funds rate alone.

Two lessons for practice follow. First, the anchor rule of Section 2.3 is further from the best choice in this panel than in the simulation reported there. The conventional specification gains more than 90 log points from the common anchor set, which was not chosen with

[10]The twelve series of the short list were left out so that the set can be used in every specification. Two of the ten replacements raised the log marginal likelihood by more than its variation across chains, by 7.6 and 11.2 log points, and the two together raised it by 15.1. The resulting set keeps household financial assets as an anchor.

it in mind. Choosing one anchor set by marginal likelihood and holding it fixed across the specifications being compared removes this source of variation at the cost of a few additional fits. Second, a search that ranks specifications by a single chain can be misled when the estimate for some specification varies across chains. The leading specifications should therefore be re-estimated with several chains before one of them is reported.

4.5 Responses to a policy shock

We identify the policy shock as the innovation in the federal funds rate that is uncorrelated with the innovations of the other observed factors in the same quarter. This amounts to ordering the funds rate last among the observed factors in a recursive scheme. Under IRc the innovations of the latent factors are uncorrelated with those of the observed factors, so the latent factors do not respond on impact, while every panel series can respond on impact through its loading on the funds rate. The shock is a linear function of the innovation v_t in y_t, so by Proposition 1 its effects are the same under every normalization of the specification. For each specification we ran a further chain of 1000 burn-in and 8000 retained draws with $r_1 = 5$, drew Γ from its conditional distribution in Lemma 2 at 1000 evenly spaced draws, and computed the responses to a shock that raises the funds rate by 25 basis points on impact. Responses of series that enter in differences are cumulated, so they refer to levels. The selected specification has posterior probability above 0.9999 among those visited, and above 0.999 among the specifications of Table 3, so averaging over specifications would not change its responses.

Figure 1 shows the responses of six series. In the selected specification the funds rate returns to zero after about three years. Industrial production falls by about 0.2 percent after two years, and the unemployment rate rises by about 0.04 percentage points. Consumer prices barely move in the first two years, with a median rise of 0.02 percent in the first year whose bands include zero, and they are about 0.2 percent lower after five years. Real M2 falls by about 0.08 percent in the first year and is back at its initial level after three years.

The responses of the selected specification and of the specification of Bernanke et al. (2005) are close. For industrial production, consumer prices and the unemployment rate their medians differ by at most 0.03 at every horizon. From two years on, their 90 percent bands are at least 0.17 wide for industrial production and consumer prices and at least 0.06 wide for the unemployment rate. For the unemployment rate the medians differ by at most 0.01. The large difference in marginal likelihood between the two specifications, 584 log points with five latent factors, therefore does not show up in the responses of these series to policy. The five series added by the search improve the description of the panel. The

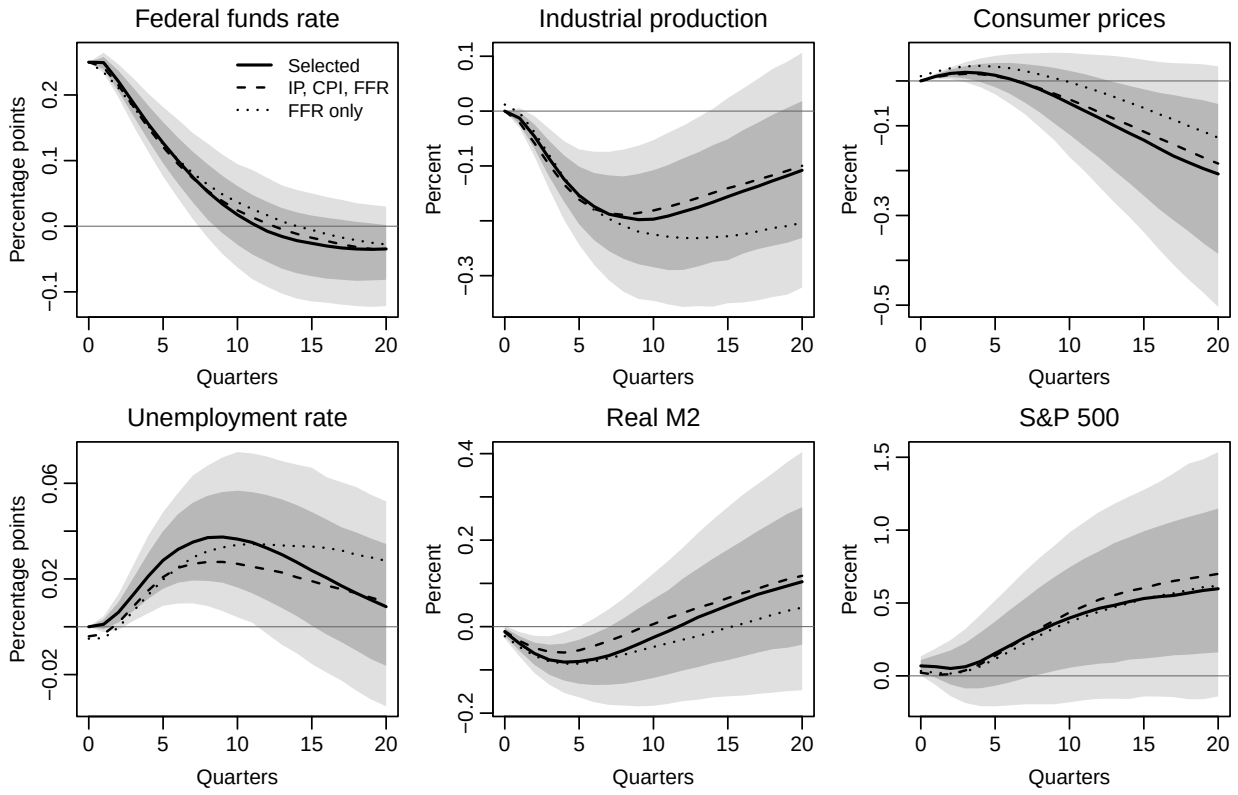


Figure 1: Responses to a 25 basis point policy shock, $r_1 = 5$. The solid line is the posterior median for the selected specification, and the shaded areas are its 68 and 90 percent pointwise credible bands. The dashed and dotted lines are the posterior medians for the specifications with industrial production, consumer prices and the federal funds rate, and with the federal funds rate alone. Industrial production, consumer prices, real M2 and the S&P 500 are in percent of their levels, and the federal funds rate and the unemployment rate in percentage points.

information they carry about the transmission of policy, however, appears to be captured already by the latent factors when they stay in the panel. The specification with the funds rate alone differs more. Its decline in industrial production is more persistent, 0.20 percent after five years against 0.11 percent, consumer prices fall less, 0.13 percent after five years against 0.21 percent, and the initial rise in prices is somewhat larger. The comparison that the marginal likelihood decides by 59 log points thus matters more for the responses than the comparison it decides by 584.

Two checks leave these conclusions unchanged. With four latent factors the medians of the selected specification move by at most 0.02 for the four macroeconomic series. Ordering household assets after the funds rate, since financial assets may respond within the quarter, moves the medians of the macroeconomic series by at most 0.03.[11]

[11]It changes the response of the S&P 500, from 0.07 to −0.08 percent on impact and from 0.60 to 0.48

5 Conclusion

The choice of observed factors in a FAVAR is usually made by convention. We have shown that it can be made by marginal likelihood, because every split of a panel describes the same data, and that the comparison is feasible for panels of realistic size once the factors and the loadings on the observed factors are integrated out. The priors are the main obstacle. A fixed prior scale on the loadings introduces a penalty that grows with the size of the panel and can reverse the ranking of splits. A hierarchical prior with a common scale estimated from the cross-section leaves a comparison that is driven by the data. The prior on the noise variances decides the closest comparison, between an observed factor and an additional latent factor anchored on the same series, and a hierarchical prior separates the two by at least ten log points in every data set of our baseline design. How strongly that prior pulls each noise variance toward the common scale sets how close to exact a series must be before it is treated as an observed factor. In panels of 100 series the procedure recovers the observed factors and the number of latent factors without either being fixed in advance. A weak observed factor is missed in about half of the data sets when the number of latent factors is free, most often because an additional latent factor can reproduce it closely.

In a monetary FAVAR estimated on FRED-QD, the procedure needs one extension, serially correlated idiosyncratic errors, and one restriction, a fixed number of latent factors. With these it selects a VAR of eight series that contains the three series of Bernanke et al. (2005), by margins of 11 and 24 log points with four and five latent factors. When the same anchor series are used in every specification, the margin with five latent factors falls to about 9 log points, and the eight series VAR is still preferred. The responses to a policy shock from the selected VAR are close to those of the conventional specification with industrial production, consumer prices and the federal funds rate, while the funds rate alone gives noticeably different responses at long horizons. The marginal likelihood therefore supports the informal choice that Bernanke et al. (2005) made between their two specifications. The additional series it favours change the description of the panel more than the estimated effects of policy.

Two questions remain open. The first is the number of latent factors in macroeconomic panels. In our application the marginal likelihood rewards every additional latent factor up to twelve, far beyond the estimate of IC_{p2}. Part of this may come from the estimator, since with eight latent factors bridge estimates from independent chains differ by up to 63 log points and short chains overstate them. Part may reflect features of the data that the

percent after five years. The S&P 500 rises in the medium run in every specification, but its 90 percent band includes zero after two and after five years, so we do not interpret this response.

model omits, such as changes in volatility over the sample, which additional factors can partly absorb. Separating the two requires estimates of the marginal likelihood that remain reliable with well over a thousand parameters, which is the second open question. Near the boundary of Proposition 2 our checks found bridge sampling stable and the Laplace–Metropolis estimator unreliable, so the cheap diagnostic we report is least informative exactly where the comparison is closest.

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Table 4: Fixed and hierarchical loading priors with one latent factor

N	Design	Prior	Correct	$\Pr(\mathcal{M}_0 \mid X)$	Margin		$\Pr(x_2 \in S)$	$\Pr(\text{decoy} \in S)$
					Median	Min		
100	Near-exact	Hier.	10/10	1.000	44.9	18.9	1.000	0.000
100	Near-exact	Fixed	9/10	0.900	47.1	−19.8	0.900	0.100
100	Weak x_2, $w = 0.3$	Hier.	10/10	1.000	132.8	113.1	1.000	0.000
100	Weak x_2, $w = 0.3$	Fixed	0/10	0.000	−71.4	−123.3	0.000	0.000
100	Weak x_2, $w = 0.2$	Hier.	7/10	0.651	12.9	−34.9	0.651	0.000
100	Weak x_2, $w = 0.2$	Fixed	0/10	0.000	−244.8	−273.9	0.000	0.000
250	Weak x_2, $w = 0.3$	Hier.	10/10	1.000	289.6	262.6	1.000	0.000
250	Weak x_2, $w = 0.3$	Fixed	6/10	0.600	32.8	−192.8	0.600	0.000

Notes. One latent factor, $r_1 = 1$ known, $T = 150$, true VAR $\{\text{policy rate}, x_1, x_2\}$, 32 candidate splits, ten data sets per row. Correct is the number of data sets in which the true specification $\mathcal{M}_0$ has the highest marginal likelihood. Margin is $\log m(X \mid \mathcal{M}_0)$ minus the largest log marginal likelihood among the other splits, so negative values mean that another split is preferred. $\Pr(\text{decoy} \in S)$ is the largest average inclusion probability among x_3, x_4, x_5. The fixed prior gives every loading a $\mathcal{N}(0, 10)$ distribution. The hierarchical prior uses $\mathcal{IG}(1/2, 1/2)$ hyperpriors.

A Additional simulation evidence

A.1 Fixed and hierarchical loading priors

The hierarchical prior on the loadings matters most when an observed factor is weak. An earlier set of simulations isolates this effect by fixing everything except the set of observed factors. The true VAR contains the policy rate, x_1, x_2 and one latent factor. The policy rate is in the VAR in every specification, $r_1 = 1$ is known, and the 32 subsets of the candidates $x_1, \ldots, x_5$ are enumerated, with $T = 150$. The decoys load on the latent factor with coefficients drawn from $U(0.6, 1.2)$, and the panel series are generated as in Section 3.1 with one latent factor. We consider the near-exact design, and the weak-factor design with $w \in \{0.3, 0.2\}$, where w multiplies the effect of x_2 on every factor and all loadings on x_2, with $N = 100$. We also consider the weak-factor design with $w = 0.3$ and $N = 250$, so that the panel has more series than periods. The hierarchical loading prior is compared with a fixed-scale prior in which all loadings are $\mathcal{N}(0, 10)$. Both use a fixed prior $\mathcal{IG}(1/2, 0.005)$ on the noise variances, and each specification is estimated with 500 burn-in and 4000 retained draws. Each cell has ten data sets, and the two priors are applied to the same data sets. Table 4 reports the results.

With $w = 0.3$ and 100 series the two priors disagree in every data set. The hierarchical prior selects the true split in all ten, with margins between 113 and 146 log points. The fixed-scale prior selects the split without x_2 in all ten, by margins between 36 and 123 log

points. The reason is the complexity penalty of Remark 1. Moving x_2 from the panel into the VAR adds a loading on x_2 for each of the 97 series that remain in the panel. Under a $\mathcal{N}(0, v)$ prior each added loading costs roughly $\frac{1}{2}\log(1 + vT/\sigma_i^2)$, which is about four log points for $v = 10$, $T = 150$ and the noise variances of this design. The penalty for the move is therefore of the order of several hundred log points, and a weak observed factor cannot improve the fit by that much. Under the hierarchical prior v is replaced by the common scale τ_Γ^2, which is estimated from the loadings and is far below ten for standardized series. The penalty per loading is then smaller, and the comparison turns on whether x_2 improves the fit.

At $w = 0.2$ the fixed-scale prior again drops x_2 in every data set, now by 207 to 274 log points. The hierarchical prior ranks the true split first in seven data sets and prefers the split without x_2 in three, by 10.7, 12.7 and 34.9 log points. With 250 series and $w = 0.3$ the hierarchical prior selects the true split in all ten data sets, with margins about twice those with 100 series, while the fixed-scale prior drops x_2 in four of them, by 25 to 193 log points. Additional series raise both the gain in fit from x_2 and the number of loadings that move with it, so under a fixed scale the outcome depends on the data set. With near-exact decoys both priors rank the true split first in nine or ten data sets, and the one exception, under the fixed prior, replaces x_1 and x_2 by decoys.

These results differ from the weak-factor results of Table 1 in an instructive way. With r_1 fixed, the alternative to placing x_2 in the VAR is to leave it in the panel as a noisy indicator of the existing factors, and the hierarchical prior rejects that alternative by more than 100 log points at $w = 0.3$. When r_1 is free, x_2 can instead be treated as the anchor of an additional latent factor, which reproduces it far more closely, and the comparison becomes the close one of Proposition 2.[12]

A.2 Sensitivity to the hyperpriors

The procedure leaves the researcher three choices in the priors of Section 2.5. They are the hyperprior (a_τ, b_τ) of the loading scales, the parameter ν_σ of the prior on the noise variances, and the hyperprior (a_σ, b_σ) of their common scale. We changed each of them in turn and re-estimated the leading specifications of the data sets in which the decision is closest. For the loading scales we used the tighter hyperprior $\mathcal{IG}(2, 2)$ and the nearly flat $\mathcal{IG}(0.05, 0.005)$ in place of $\mathcal{IG}(1/2, 1/2)$. For the noise variances we used $\nu_\sigma = 4$ in place of $\nu_\sigma = 1$, which pulls each noise variance more strongly toward the common scale s_σ, and the

[12] The results with 100 series in Table 4 were computed before Step 8 of Algorithm 1 was added. Re-estimating the leading specifications of three data sets per cell with Step 8 left the selected split unchanged in all 18 data sets and changed the margin by at most 3.7 log points.

Table 5: Sensitivity to the hyperpriors, $N = 100$

		Noise variances		Loading scales	
Data sets	Default	$\nu_\sigma = 4$	$s_\sigma \sim \mathcal{G}(0.1, 0.1)$	$\mathcal{IG}(2, 2)$	$\mathcal{IG}(0.05, 0.005)$
Baseline, 2 closest	2/2	2/2	2/2	2/2	2/2
Near-exact, 2 closest	1/2	0/2	1/2	1/2	1/2
Near-exact, all 20	19/20	14/20			
Weak x_2, 12 closest	3/12	3/12	3/12	3/12	1/12

Notes. Number of data sets in which the true specification has the highest log marginal likelihood among the true specification and the three best other specifications of the original search, each re-estimated with 1000 burn-in and 16000 retained draws under the stated prior. Default is the prior of Section 3.1, with $\nu_\sigma = 1$, $s_\sigma \sim \mathcal{G}(1, 1)$ and loading scales $\mathcal{IG}(1/2, 1/2)$. Each other column changes one of these. The closest data sets are those with the smallest margins in Table 1, including all data sets in which the truth was not selected.

diffuse hyperprior $s_\sigma \sim \mathcal{G}(0.1, 0.1)$ in place of $\mathcal{G}(1, 1)$. The data sets are the two baseline data sets with the smallest margins, the near-exact data set in which a decoy was selected and the one with the smallest margin, and the nine weak-factor data sets in which the truth was not selected together with the three in which it won by the smallest margins. In each data set we re-estimated the true specification and the three best other specifications of the original search with long chains. Because ν_σ turned out to matter, we repeated this step for all 20 near-exact data sets under $\nu_\sigma = 4$. Table 5 reports the number of data sets in which the true specification is the best of those re-estimated.

The diffuse hyperprior on s_σ leaves the results unchanged. Every margin is within 0.4 log points of its default value, as expected, since s_σ is estimated from the noise variances of more than ninety series. The hyperprior on the loading scales leaves the baseline and near-exact margins within 1.6 log points of their default values, but it matters in the weak-factor design, for a reason we explain below.

The parameter ν_σ matters, and in the direction that Remark 2 predicts. With $\nu_\sigma = 4$ the prior density of each σ_i^2 falls off faster below the common scale, so a specification that treats a nearly exact series as the anchor of a latent factor pays a larger price. Genuine observed factors gain from this. The margins of the two baseline data sets rise from 10.1 and 11.5 to 26.6 and 32.2 log points, and in the weak-factor design the margin in the data set where the truth had won by 2.8 rises to 17.8. Decoys gain in the same way. Among the 20 near-exact data sets the true specification is selected in 14, against 19 under the default prior. In all six exceptions the selected specification puts a decoy in the VAR in place of a latent factor, with the truth between 6.2 and 40.9 log points behind. The parameter ν_σ therefore sets how small the noise variance of a series must be, relative to that of the rest of the panel, before the series is treated as an observed factor. The default $\nu_\sigma = 1$ is a weak pull toward the

common scale and leaves this threshold mostly to the data. A researcher who regards series with a noise share of one percent as factors measured without error can raise ν_σ, and should then expect such series to enter the VAR.

The hyperprior on the loading scales matters in the weak-factor design because one competing specification has an unusual loading scale. Under the nearly flat $\mathcal{IG}(0.05, 0.005)$, four of the twelve data sets select x_2 alone in the VAR with five latent factors, with the truth 18 to 34 log points behind. In this specification the loadings on the observed factors are the loadings on x_2 alone, which are small by construction, so their common scale τ_Γ^2 is small as well. The default $\mathcal{IG}(1/2, 1/2)$ penalizes such small scales heavily, while $\mathcal{IG}(0.05, 0.005)$ puts much more mass on them.[13] The hyperprior thus decides the comparison only when a specification contains observed factors whose loadings are all much smaller than those of the other factors. Hyperpriors that are nearly flat in this sense are known to be informative about small scales (Gelman, 2006).

These comparisons re-estimate the leading specifications of the original search and do not repeat the search under each prior. A prior that favours a specification the original search did not rank highly could lead to a different selection, so the counts in Table 5 for the alternative priors are upper bounds on the number of correct selections.

B Proofs

Proof of Proposition 1. For any A of the stated form, $\Lambda^* f_t^* + \Gamma^* y_t = \Lambda M_{11}^{-1}(M_{11} f_t + M_{12} y_t) + (\Gamma - \Lambda M_{11}^{-1} M_{12}) y_t = \Lambda f_t + \Gamma y_t$, so the measurement equation and the common component are unchanged. Premultiplying (2) by A gives $s_t^* = \sum_p A B_p A^{-1} s_{t-p}^* + A u_t$ with $\mathrm{Var}(A u_t) = A \Omega A'$, and the initial states transform in the same way. Since $|A| = |M_{11}|$ is constant, the joint distribution of (Z, Y) is unchanged. With $M_{11} = I_{r_1}$ and $M_{12} = \Gamma_1$, the transformed loadings satisfy $\Lambda_1^* = \Lambda_1 = I_{r_1}$ and $\Gamma_1^* = \Gamma_1 - \Lambda_1 \Gamma_1 = 0$, which is BBE. Conversely, the upper right block of $A \Omega A'$ with $M_{11} = I_{r_1}$ is $\Omega_{fy} + M_{12} \Omega_{yy}$, which vanishes for $M_{12} = -\Omega_{fy} \Omega_{yy}^{-1}$, while $\Lambda_1^* = \Lambda_1 = I_{r_1}$. Applying the first map to an IRc parameter gives $\Omega_{fy}^{\mathrm{BBE}} = \Gamma_1 \Omega_{yy}$ and $\Omega_{yy}^{\mathrm{BBE}} = \Omega_{yy}$, so the second map uses $M_{12} = -\Gamma_1$ and $A_{-\Gamma_1} A_{\Gamma_1} = I$. The two maps are therefore inverse to each other.

For the responses, the last r_2 rows of A equal $(0, I_{r_2})$, so the innovation to y_t is the same

[13] In one data set the posterior median of τ_Γ^2 is 0.021 in this specification, against 0.097 in the true one. Moving from the default to the nearly flat hyperprior raised the log marginal likelihood of the true specification by 1.0 log point and that of the alternative by 31.9, and reweighting the draws of the scales by the ratio of the two hyperpriors accounts for most of the second change. Under the tighter $\mathcal{IG}(2, 2)$ the margin of the truth over this specification rises by 14.2 and 11.2 log points in the two data sets where it was the main competitor.

in every parameterization, and the information sets generated by $\{s_{t-j}\}$ and $\{s^*_{t-j}\}$ coincide because A is invertible. Under Gaussianity $\mathrm{E}[u_t \mid v_t] = \Omega_{\cdot y}\Omega_{yy}^{-1} v_t$, where $\Omega_{\cdot y}$ denotes the last r_2 columns of Ω. Hence $R_h = L\Psi_h\Omega_{\cdot y}\Omega_{yy}^{-1}$, where Ψ_h are the moving average coefficients of (2) and L stacks the loadings (Λ, Γ) above $(0, I_{r_2})$. Under the transformation, $L^* = LA^{-1}$, $\Psi^*_h = A\Psi_h A^{-1}$ and $\Omega^*_{\cdot y} = A\Omega A'(0, I_{r_2})' = A\Omega_{\cdot y}$, because $A'(0, I_{r_2})' = (0, I_{r_2})'$. The product is therefore unchanged. □

Proof of Lemma 1. The vector Hs contains the initial state and the innovations, which are independent with block diagonal covariance $\mathcal{V}$, so $s = H^{-1}(Hs)$ is Gaussian with precision $H'\mathcal{V}^{-1}H$. Because H has nonzero blocks only on the diagonal and the first P subdiagonals, K has nonzero blocks only within P blocks of the diagonal, and so does any submatrix that keeps the same coordinates in each time block. The conditional distribution of f given y follows from the standard formula for a Gaussian vector in precision form. For the full conditional, the log of $p(Z \mid f, y, \theta)\, p(f \mid y, \theta)$ equals, up to terms that do not involve f,

$$
\begin{aligned}
&-\frac{1}{2}\big(\tilde{z} - (I_T \otimes \Lambda)f\big)' R \big(\tilde{z} - (I_T \otimes \Lambda)f\big) - \frac{1}{2}(f - \mu_f)' K_{ff} (f - \mu_f) \\
&\qquad = -\frac{1}{2} f' Q f + f' \left(K_{ff}\mu_f + (I_T \otimes \Lambda')R\tilde{z}\right) + \text{const.}
\end{aligned}
$$

The matrix $(I_T \otimes \Lambda')R(I_T \otimes \Lambda)$ has the block structure of R, block diagonal in the baseline model and block tridiagonal under (3). Since $P \geq 1$, Q inherits the band structure of K_{ff} in both cases, and the cost statement is the cost of a banded Cholesky factorization with bandwidth Pr_1. □

Proof of Lemma 2. Stacking (1) gives $z = (Y \otimes I_n)\gamma + (I_T \otimes \Lambda)f + \varepsilon$. Integrating over $f \sim \mathcal{N}(\mu_f, K_{ff}^{-1})$ gives $z \sim \mathcal{N}((Y \otimes I_n)\gamma + (I_T \otimes \Lambda)\mu_f,\, W)$ with $W = R^{-1} + (I_T \otimes \Lambda)K_{ff}^{-1}(I_T \otimes \Lambda')$. The Woodbury identity gives the stated W^{-1}, since $K_{ff} + (I_T \otimes \Lambda')R(I_T \otimes \Lambda) = Q$. The posterior of γ is that of a Gaussian linear regression with known error covariance W. In the baseline model, $R = I_T \otimes \Sigma^{-1}$ and the expression for D^{-1} uses $(Y' \otimes I_n)(I_T \otimes \Sigma^{-1})(Y \otimes I_n) = Y'Y \otimes \Sigma^{-1}$ and $(Y' \otimes I_n)(I_T \otimes \Sigma^{-1}\Lambda) = Y' \otimes \Sigma^{-1}\Lambda$. □

Proof of Proposition 3. With the other parameters fixed and a flat prior on $\gamma = \mathrm{vec}(\Gamma)$, the joint conditional of (f, γ) is Gaussian with precision blocks $\mathcal{P}_{ff} = K_{ff} + I_T \otimes G$, $\mathcal{P}_{f\gamma} = Y \otimes \Lambda'\Sigma^{-1}$ and $\mathcal{P}_{\gamma\gamma} = Y'Y \otimes \Sigma^{-1}$, where the cross term follows from $f_t'\Lambda'\Sigma^{-1}\Gamma y_t = f_t'(y_t' \otimes \Lambda'\Sigma^{-1})\gamma$. For a two-block Gibbs sampler on a Gaussian target, the draws of γ form a Gaussian autoregression with coefficient matrix $\mathcal{P}_{\gamma\gamma}^{-1}\mathcal{P}_{\gamma f}\mathcal{P}_{ff}^{-1}\mathcal{P}_{f\gamma}$, whose spectral radius is the geometric rate of convergence and the lag-one autocorrelation of the slowest linear function (Amit, 1991; Liu et al., 1994; Roberts and Sahu, 1997). Its nonzero eigenvalues coincide with

those of $\mathcal{P}_{ff}^{-1}\mathcal{P}_{f\gamma}\mathcal{P}_{\gamma\gamma}^{-1}\mathcal{P}_{\gamma f}$, and

$$\mathcal{P}_{f\gamma}\mathcal{P}_{\gamma\gamma}^{-1}\mathcal{P}_{\gamma f} = (Y \otimes \Lambda'\Sigma^{-1})\left((Y'Y)^{-1} \otimes \Sigma\right)(Y' \otimes \Sigma^{-1}\Lambda) = P_Y \otimes G,$$

which gives the expression for ϱ^2. Because $\mathcal{P}_{ff}$ is positive definite and $P_Y \otimes G$ is positive semidefinite, $\varrho^2 = \max_x x'(P_Y \otimes G)x/x'\mathcal{P}_{ff}x$. Take $x = Ya \otimes w$ for any nonzero $a \in \mathbb{R}^{r_2}$ and a unit eigenvector w of G with eigenvalue g. Then $x'(P_Y \otimes G)x = x'(I_T \otimes G)x = g\,a'Y'Ya$ and $x'K_{ff}x \le \kappa\,a'Y'Ya$, which gives $\varrho^2 \ge g/(g+\kappa)$. Finally, drawing γ from its marginal and then f from its conditional given γ is a draw from the joint Gaussian distribution of (f, γ) that does not depend on the previous state. □

Proof of Proposition 4. Let $\psi = (\Lambda, \Sigma, B, \Omega, \tau_\Lambda^2, \tau_\Gamma^2, s_\sigma)$ and suppose the current state (f, Γ, ψ) is distributed according to the posterior. Then ψ is distributed according to its marginal posterior. Step 1 draws Γ from $\pi(\Gamma \mid \psi, Z, y)$, which does not involve the current f or Γ, and Step 2 draws f from $\pi(f \mid \Gamma, \psi, Z, y)$. Together they draw (Γ, f) from its joint conditional given ψ, so after Step 2 the state again has the posterior distribution. When $r_2 = 0$, Step 1 is absent and Step 2 is a draw from the full conditional of f. Steps 3 to 7 are draws from full conditional distributions and each preserves the posterior. In Step 5 the three scales are conditionally independent given the other parameters, since τ_Λ^2, τ_Γ^2 and s_σ enter the posterior only through the priors of Λ_2, Γ and Σ respectively, and the full conditional of s_σ is the stated gamma distribution because the σ_i^2 are conditionally independent inverse gamma variables with rate $s_\sigma/2$. Under (3), the update of each ρ_i in Step 4 is a Metropolis–Hastings step whose target is the full conditional of ρ_i, because the proposal density times the acceptance ratio reproduces the factor $(1-\rho_i^2)^{1/2}$ of that conditional, and it therefore preserves the posterior.

For Step 8, write x for the state $(f, \theta, \tau_\Lambda^2, \tau_\Gamma^2, s_\sigma)$ and g_c for the map (6). The maps form a group under composition, $g_{c'} \circ g_c = g_{c'c}$ with componentwise products, and $g_{1/c}$ inverts g_c. Each g_c maps the IRc parameter space onto itself, because A_c is block diagonal and leaves Λ_1 and the zero block Ω_{fy} unchanged, and it is linear in x with constant Jacobian $J_c = \prod_k c_k^{T-(n-r_1)+r_1+1}$, so that $J_{1/c} = 1/J_c$. The proposal density of c with respect to the Haar measure $\prod_k dc_k/c_k$ is that of $\log c$ with respect to Lebesgue measure, and it is the same at c and $1/c$. The Haar measure is invariant under $c \mapsto 1/c$. For a state x and the proposal $x' = g_c(x)$, the reverse proposal is $x = g_{1/c}(x')$, and with $\alpha(x, c) = \min\{1, \pi(g_c(x))J_c/\pi(x)\}$,

$$\pi(x)\,\alpha(x,c) = \min\{\pi(x), \pi(x')J_c\} = J_c\,\min\{\pi(x'), \pi(x)J_{1/c}\} = J_c\,\pi(x')\,\alpha(x', 1/c).$$

Since $dx' = J_c\,dx$, this is detailed balance with respect to $\pi(x)\,dx$, and Step 8 preserves the

posterior (Liu and Sabatti, 2000). Under (3) the map leaves ρ unchanged, so the Jacobian and the argument are the same. The composition of all steps therefore preserves the posterior. □

Proof of Proposition 2. Order the observed factors of $\mathcal{M}$ as $(y_S', y_j)'$ and the states as $s_t = (f_t', y_{St}', y_{jt})'$, and partition Ω_{yy} conformably into Ω_{SS}, Ω_{Sj} and Ω_{jj}. Let $C = \Omega_{jS}\Omega_{SS}^{-1}$, with C absent if S is empty, and define $g_t = y_{jt} - Cy_{St}$. The map $s_t \mapsto s_t' = (f_t', g_t, y_{St}')'$ is linear and invertible, so as in the proof of Proposition 1 the vector s_t' follows a VAR(P) with coefficients AB_pA^{-1} and innovation covariance $A\Omega A'$, where A is the matrix of the map. The innovation of g_t is $u_{jt} - Cu_{St}$, whose covariance with u_{St} is $\Omega_{jS} - C\Omega_{SS} = 0$, and the innovations of f_t are uncorrelated with those of y_{St} by IRc. Hence the latent states $(f_t', g_t)'$ have innovations uncorrelated with those of the observed states y_{St}, as IRc requires in $\mathcal{M}'$.

In $\mathcal{M}$, series j is $X_{jt} = y_{jt} = g_t + Cy_{St}$, and every panel series satisfies $X_{it} = \Lambda_i f_t + \Gamma_{iS}y_{St} + \Gamma_{ij}y_{jt} + \varepsilon_{it} = \Lambda_i f_t + \Gamma_{ij}g_t + (\Gamma_{iS} + \Gamma_{ij}C)y_{St} + \varepsilon_{it}$. These are the measurement equations of $\mathcal{M}'$ with latent factors $(f_t', g_t)'$, loadings (Λ_i, Γ_{ij}) on them, loadings $\Gamma_{iS} + \Gamma_{ij}C$ on y_S, and, for series j, loadings $(0, 1)$ and C with noise variance zero. With the anchors of $\mathcal{M}$ followed by series j, the anchor block of the latent loadings is

$$\begin{pmatrix} I_{r_1} & \Gamma_{1j} \\ 0 & 1 \end{pmatrix},$$

where Γ_{1j} collects the loadings of the anchors of $\mathcal{M}$ on y_j. It is invertible, and the transformation of Section 2.3 with M_{11} equal to this block and $M_{12} = 0$ sets it to the identity without changing the distribution of the data. This transformation acts only on the latent states, so the innovations of the latent states remain uncorrelated with those of y_S. Call the resulting parameter θ_0'. It has the structure of $\mathcal{M}'$ and satisfies IRc except that $\sigma_j^2 = 0$, and under θ_0' the data have the same distribution as under θ.

For $\epsilon > 0$, θ_ϵ' is a parameter of $\mathcal{M}'$. Under θ_ϵ' the vector X is Gaussian with mean zero and covariance $V_\epsilon = V_0^X + \epsilon\, E_j$, where V_0^X is its covariance under θ and E_j is the diagonal matrix with ones in the positions of $X_{j1}, \ldots, X_{jT}$. Since $p(X \mid \theta, \mathcal{M})$ is a Gaussian density, V_0^X is positive definite, and the Gaussian density is continuous in its covariance on the set of positive definite matrices, so $p(X \mid \theta_\epsilon', \mathcal{M}') \to p(X \mid \theta, \mathcal{M})$ for every X. Under (3), the other series keep their coefficients ρ_i, and setting $\rho_j = 0$ in θ_0' gives the same covariance V_ϵ, so the argument is unchanged. □

Proof of Proposition 5. Since IC_{p2} chooses among at most $k_{\max} + 1$ values, $k = \hat{r} + 2$ takes values in a finite set $\mathcal{K}$. On the event $\{\hat{r} \geq r - 2\}$, whose probability tends to one by (i), $k \geq r$, and then $\max_j |\hat{u}_j(k) - u_j^*| \leq \delta_N = \max_{k' \in \mathcal{K},\, k' \geq r} \max_j |\hat{u}_j(k') - u_j^*|$, which tends to zero in probability by (ii) because $\mathcal{K}$ is finite. On the event $\{\delta_N < c/2\}$, every $j \in S_0$ has

$\hat{u}_j(k) < c/2$ and every $j \notin S_0$ has $\hat{u}_j(k) > c/2$ by (iii). The r_2 smallest values of $\hat{u}_j(k)$ then belong to the series of S_0, which are therefore among the $J \geq r_2$ series with the smallest $\hat{u}_j$. Both events have probability tending to one. □

Proof of Lemma 3. For (a), the identity $p(Z \mid y, \theta) = p(Z \mid f, y, \theta)p(f \mid y, \theta)/p(f \mid Z, y, \theta)$ holds for every f. At $f = \hat{f}$ the denominator equals $(2\pi)^{-Tr_1/2}|Q|^{1/2}$ and the prior density equals $(2\pi)^{-Tr_1/2}|K_{ff}|^{1/2}\exp\{-\frac{1}{2}(\hat{f}-\mu_f)'K_{ff}(\hat{f}-\mu_f)\}$, which gives the first expression. The marginal distribution of y under $s \sim \mathcal{N}(0, K^{-1})$ has precision equal to the Schur complement $K_{yy} - K_{yf}K_{ff}^{-1}K_{fy}$, whose log determinant is $\log|K| - \log|K_{ff}|$, which gives the second. For (b), the identity $p(Z, y \mid \phi) = p(Z, y \mid \phi, \gamma)\,\pi(\gamma \mid \phi)/\pi(\gamma \mid Z, y, \phi)$ holds for every γ, and the posterior of γ given ϕ is $\mathcal{N}(\hat{\gamma}, D)$ by Lemma 2, because $p(y \mid \theta)$ does not depend on Γ. □

C The sampler with autoregressive errors

Under (3), Steps 1 and 2 use the precision R in Lemmas 1 and 2. In Steps 3 and 4 the regression of series i on the factors is transformed with ρ_i as in Section 2.2, so that its errors are independent with variance σ_i^2, and the full conditionals keep their Gaussian and inverse gamma forms. Given the errors $\varepsilon_{i1}, \ldots, \varepsilon_{iT}$ and σ_i^2, the full conditional density of ρ_i on $(-1, 1)$ is proportional to $(1-\rho_i^2)^{1/2}$, the contribution of the stationary distribution of ε_{i1}, times a Gaussian density with precision $h_i = \sum_{t=2}^{T-1}\varepsilon_{it}^2/\sigma_i^2 + 1/0.5^2$ and mean $h_i^{-1}\sum_{t=2}^{T}\varepsilon_{it}\varepsilon_{i,t-1}/\sigma_i^2$. We draw ρ_i by an independence Metropolis–Hastings step whose proposal is this Gaussian density truncated to $(-1, 1)$, so that a proposal ρ_i' is accepted with probability $\min\{1, (1-\rho_i'^2)^{1/2}/(1-\rho_i^2)^{1/2}\}$. Step 8 leaves ρ unchanged, and the anchor equations in its acceptance ratio are transformed in the same way as in Step 3.